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REVIEW ARTICLE

Computational drug design in the artificial intelligence era: A systematic review of molecular representations, generative architectures, and performance assessment

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ABSTRACT

Generative drug design has emerged as a transformative approach in pharmaceutical research, leveraging deep learning models to create novel molecules with targeted properties. This systematic review analyzes the current landscape of computational approaches across 3 critical dimensions: molecular representation strategies (1-dimensional, 2-dimensional, and 3-dimensional), generative

architectural frameworks (including variational autoencoders, generative adversarial networks, reinforcement learning systems, and diffusion models), and evaluation methodologies. We provide a comprehensive categorization of these approaches, critically assessing their relative advantages and limitations. Additionally, we examine the datasets driving development in this field and the metrics employed to evaluate model performance. Through structured analysis of these interconnected components, we identify significant research gaps and propose promising future directions for advancing artificial intelligence–driven drug discovery. This review offers researchers a unified framework for understanding the complex interplay between representation choices, generative mechanisms, and evaluation paradigms in computational drug design.

Significance Statement: This review provides a unique framework for generative drug design by categorizing methods first by drug representation and then by generative model type. This novel taxonomy clarifies which models are best suited for specific molecular data types, offering practical guidance for researchers. We also critically discuss the advantages, disadvantages, key datasets, and evaluation metrics, delivering a comprehensive and actionable resource for the field.

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I. Introduction

Drug design is the process of discovering and developing new therapeutic compounds to address diseases and medical conditions.[1–3] In recent years, computational approaches have gained significant prominence in this field by enabling researchers to simulate and predict the behavior of potential drug candidates before laboratory synthesis and testing.[4,5] This computational prescreening significantly reduces the time and resources required in early-stage drug development.[6]

The computational drug design paradigm encompasses diverse mathematical and computer-based methodologies for modeling and analyzing molecular structures and interactions.[7] These include sophisticated techniques such as molecular dynamics simulations, quantum mechanics calculations, and, increasingly, machine learning algorithms. By applying these computational tools to extensive datasets of established drugs and their targets, researchers can identify crucial patterns and relationships that guide the design of novel compounds with tailored properties.

Recent literature has extensively reviewed various aspects of computational drug design. Guo et al[8] provided a comprehensive analysis of diffusion models and their applications in bioinformatics, focusing on protein design, small molecule generation, and protein-ligand interaction modeling. Their work detailed 3 fundamental frameworks in diffusion models: denoising diffusion probabilistic models (DFPMs), noise-conditioned scoring networks, and score stochastic differential equations. Martinelli[9] conducted a systematic review of research and review papers published between January 2017 and January 2022, offering a chronological perspective on developments in the field.

Thomas et al[10] examined approaches incorporating protein target structural knowledge into de novo molecular design, categorizing them into 4 distinct groups: (1) structure-explicit distribution learning, where the protein structure and ligand information are jointly processed to learn distributions applicable to de novo molecule generation; (2) structure-implicit distribution learning, utilizing 3-dimensional (3D) bioactive conformations of bound ligands to learn distributions; (3) structure-explicit goal-directed optimization; and (4) structure-implicit goal-directed optimization, both leveraging protein representations to develop scoring functions.

Complementing these perspectives, Baillif et al[11] reviewed generative approaches for creating drug molecules in 3D representation, either through single-step generation or iterative addition of atoms or fragments. Although these approaches incorporate auxiliary information such as target structural representations to enhance reliability, they generally demonstrate lower efficiency compared with models generating molecules in 1-dimensional/2-dimensional (1D/2D) representations. Sousa et al.[12] analyzed deep generative models for targeted compound design, identifying 2 key components: the generative module for molecule creation and the biasing/exploring module that guides generation toward desired objectives. Anstine and Isayev[13] examined approaches based on variational autoencoders (VAEs), generative adversarial networks (GANs), flow models, and diffusion models, highlighting their distinctive characteristics.

The existing reviews primarily categorize approaches based on input representation, generative model architecture, or auxiliary knowledge integration. Our paper presents a novel categorization framework based on molecular representation (linear, 2D, and 3D), with subcategorization according to the employed generative method (encoder-decoder, GAN, or diffusion-based models). We analyze the temporal distribution of these approaches across the 3 representation types and evaluate their applicability to different representation frameworks. The key advantage of our representation-based taxonomy is that it highlights critical tradeoffs in how data are structured for machine learning. For instance, linear representations (eg, Simplified Molecular Input Line Entry System (SMILES) strings or sequence-based encodings) are computationally efficient but lose spatial and topological relationships. In contrast, 2D graph representations (eg, molecular graphs) preserve connectivity but may struggle with depth-aware tasks, whereas 3D representations (eg, point clouds or volumetric grids) offer high biological fidelity at the cost of increased computational complexity. By structuring our review around these categories, we provide a clearer understanding of why certain methods succeed or fail in specific applications, rather than merely listing which algorithms perform best. Our discussion sections in each representation category explicitly compare how different algorithms perform within the same input framework, revealing that algorithmic differences alone do not explain performance variations—data encoding plays a dominant role. Additionally, we review commonly used datasets and assessment metrics, analyzing their respective strengths and limitations. The paper

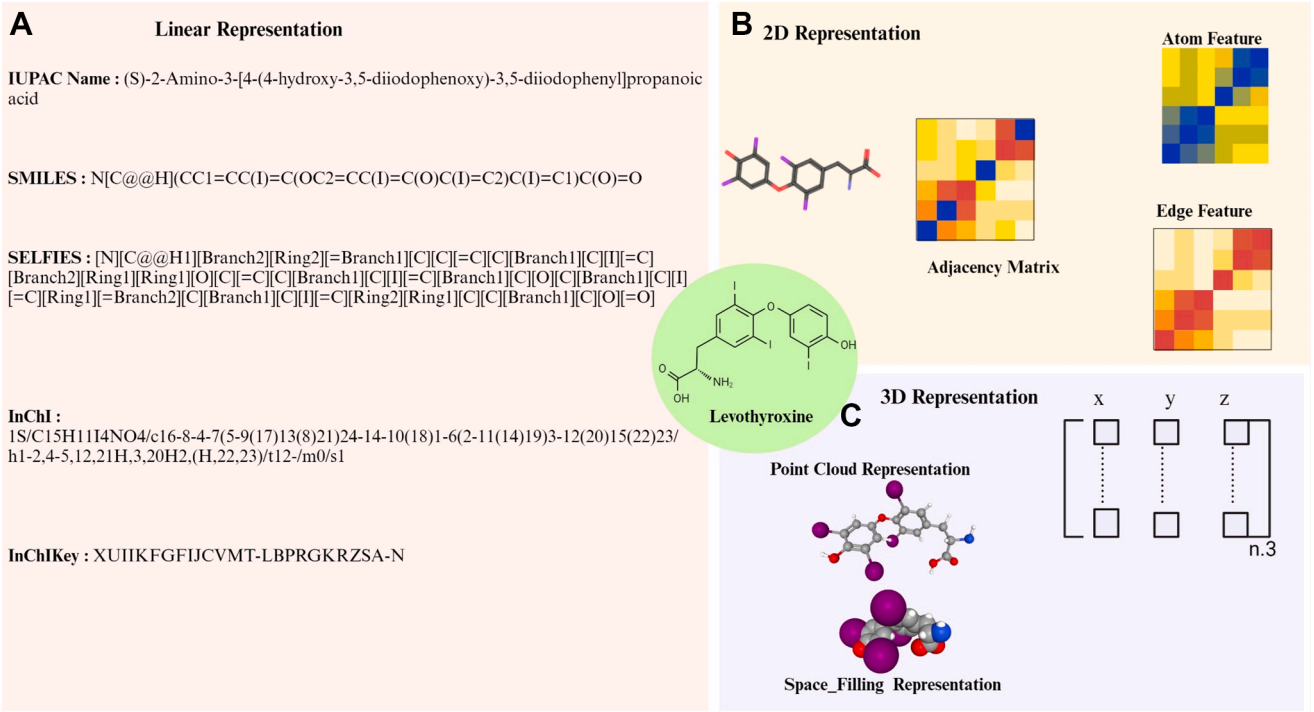


Fig. 1. A representation of a drug molecule in 3 different ways, including (A) linear representation, (B) 2D representation, and (C) 3D representation. IUPAC, International Union of Pure and Applied Chemistry.

concludes with a discussion of current methodologies and future research directions in generative drug design.

In the following, we have rigorously structured the Preferred Reporting Items for Systematic Reviews and Meta-Analyses—compliant search strategy in our review process to ensure full transparency. Our systematic search focused on generative artificial intelligence (AI) applications in molecular design (2015—2024), encompassing 7 major databases, including PubMed, arXiv, Oxford University Press, and publisher platforms from American Chemical Society, Elsevier, and Springer. The search employed carefully designed Boolean operators combining key terms such as "generative AI/deep learning" with "drug discovery/molecular design" and specific representation types (linear/2D/3D) and model architectures (VAEs/GANs/diffusion). Our inclusion criteria prioritized peer-reviewed studies and validated preprints focusing on generative approaches for small molecules or proteins while excluding nongenerative methods, purely theoretical work, and non-English publications.

II. Types of representation

Molecular representation is fundamental to computational drug design, with each representation method offering unique advantages for different applications. Drug molecules can be represented in 3 primary forms: (1) linear representation, (2) 2D representation, and (3) 3D representation. Each approach presents distinct benefits and limitations for computational analysis. Figure 1 illustrates the 3 representation types.

A. Linear representation

Linear representation provides a simplified, 1D depiction of drug molecules that effectively highlights the chemical structure and properties. This approach has become a standard technique in medicinal chemistry and drug discovery, enabling complex molecular structures to be represented in a more accessible and interpretable format.[14,15] The most prevalent linear representation is the SMILES, which uses character strings to describe molecular structure by encoding the connectivity of atoms and functional groups.[16]

Despite its widespread adoption, SMILES has limitations when representing complex molecular structures. To address these challenges, Krenn et al[17] developed Self-Referencing Embedded Strings (SELFIES) as a more robust and efficient molecular representation system. SELFIES represent molecules as sequences of embedded strings, with each string corresponding to a specific molecular fragment. This approach offers several significant advantages over traditional methods:

1. Enhanced accuracy in identifying complex molecules with multiple atoms and bonds.
2. Computational efficiency, even for large molecular structures.
3. Adaptability across various applications, including property prediction, virtual screening, and quantum chemistry calculations.

The versatility of SELFIES has been demonstrated across numerous applications in molecular property prediction, design generation, and identification of drug-like compounds, representing a significant advancement in molecular representation methodology.

Fragment-based graph generation methods offer an alternative to SMILES-based approaches by directly manipulating molecular graphs using fragments as building blocks. This can involve techniques like junction trees, where a molecule is represented as a tree of connected fragments, or hierarchical graph representations that capture relationships between fragments at different levels of detail. t-SMILES[18] utilizes a tree-based SMILES representation, whereas FragFM[19] employs a fragment-based discrete flow matching framework. These methods aim to improve molecular

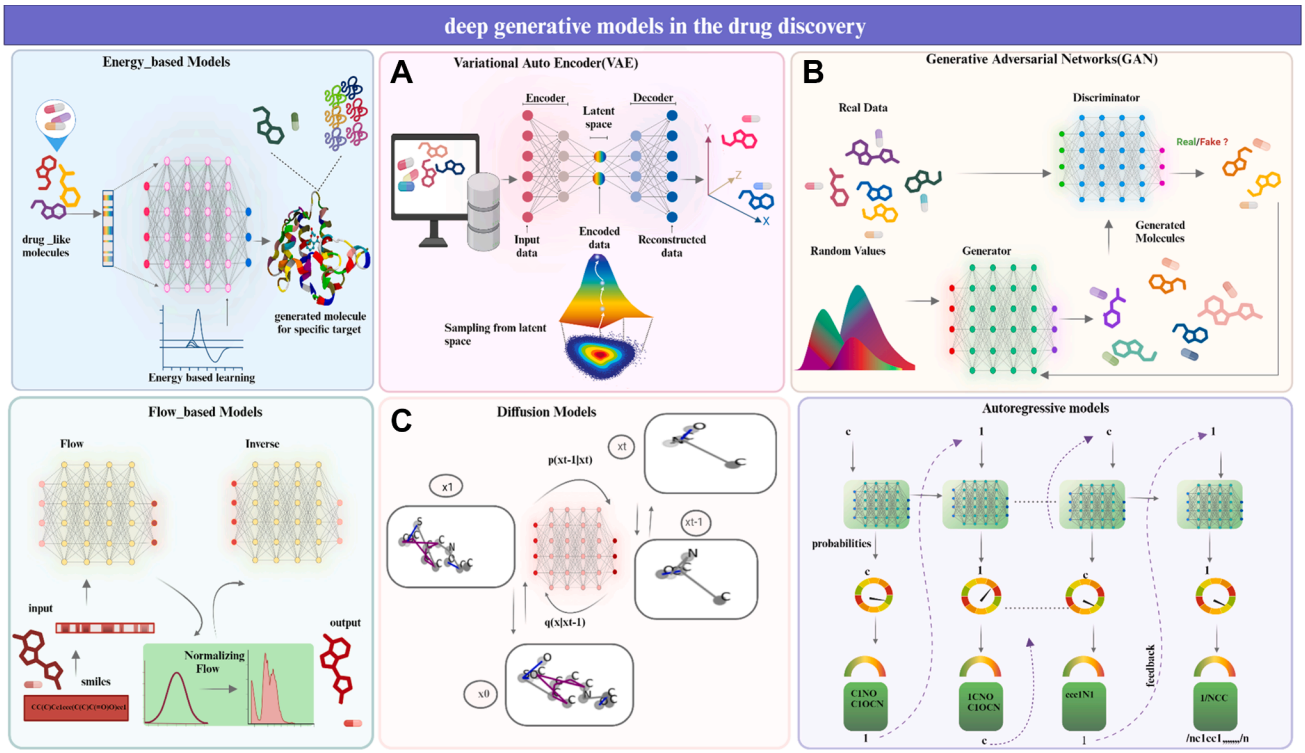


Fig. 2. The overall schematic of 3 different generative models, including (A) VAE, (B) GAN, and (C) denoising diffusion model.

validity, diversity, and efficiency in generative models, particularly in drug discovery applications.

Byte pair encoding (BPE) is a tokenization technique utilized in autoregressive models, including applications like SMILES. Autoregressive tokenizers, such as BPE for SMILES strings, aim to reduce the generation of invalid token sequences by learning meaningful subword units from a corpus of valid sequences. SMILES-pair encoding,[20] a BPE variant, consolidates frequently appearing SMILES segments into single tokens. This process aims to reduce the occurrence of invalid token sequences by recognizing and grouping together meaningful chemical substructures. Although the search results highlight BPE and its variants, the specifics of how these tokenizers inherently reduce invalid sequences aren't detailed. The key is in the training and how often certain combinations of tokens co-occur during the processing of chemical structures.

Syntax-directed VAEs (SD-VAEs) are grammar-constrained models that ensure syntactic correctness during text generation.[21] They incorporate syntactic and semantic constraints to improve the quality of generated text compared with models without these constraints. SD-VAEs achieve this by utilizing a syntax-directed generative mechanism in the decoder, which limits the output space. This approach helps produce text that is both syntactically and semantically sound. Dai et al[21] utilize SD-VAE for molecule generation.

B. 2-Dimensional graph representation

In a 2D graph representation, molecules are conceptualized as networks where atoms serve as nodes and chemical bonds function as edges. This representation preserves the topological structure of the molecule while incorporating crucial attributes for both nodes (atomic number and element type) and edges (bond order and bond type). The structured format allows researchers to visualize and analyze the relationships between different molecular components with relative ease.[22]

Advanced implementations of 2D graph representation include weighted molecular graphs, which emphasize the relative strength or importance of specific bonds through numerical weighting. Heterogeneous graphs provide additional sophistication by accommodating multiple types of nodes and edges, enabling the capture of complex interactions between various molecular components.

Mathematically, 2D molecular graphs are typically represented using 2 principal matrices: the node feature matrix and the adjacency matrix.[22] The node feature matrix contains rows corresponding to each atom, with each row encoding the specific attributes of that atom. The adjacency matrix, an $N \times N$ structure (where N represents the number of atoms), indicates connections between atoms, with the element at position (i,j) denoting the presence or absence of a bond between atoms i and j.

C. 3-Dimensional representation

3D representation provides the most comprehensive molecular visualization by displaying the spatial arrangement of atoms and bonds in physical space. This approach is particularly valuable for understanding molecular shape and geometry, critical factors that influence drug-target interactions and metabolic processes within biological systems.[23]

The traditional approach to 3D representation focuses primarily on storing the spatial coordinates of atoms, with less emphasis on bond properties. However, recent advancements have expanded to include bond characteristics in 3D models, providing more complete structural information.[24]

A specialized application of 3D representation involves the conceptual division of molecules into arms and scaffolds—a particularly useful framework in drug discovery. In this paradigm, arms primarily facilitate binding and interaction with biological

targets, whereas scaffolds provide structural support and organization for the molecule. Scaffolds typically comprise secondary structural elements (such as alpha helices and beta sheets in proteins) and create stable platforms for attaching arms, loops, and other functional groups. This organizational structure helps maintain the molecule's tertiary configuration and plays a crucial role in organizing its chemical reactivity profile.

III. Generative models

Deep generative modeling has revolutionized computational drug design, enabling the creation of novel molecular structures with desired properties. Three dominant paradigms have emerged in this space: encoder-decoder architectures, GANs, and diffusion models. Figure 2 illustrates these approaches.

A. Encoder-decoder

VAE represents a pioneering generative approach based on an encoder-decoder architecture first introduced by Kingma and Welling.[25] The innovative aspect of VAEs is their application of variational inference techniques to approximate posterior distributions over latent variables, demonstrating remarkable effectiveness in learning disentangled representations and generating novel data samples (Fig. 2A).

The VAE encoder learns to estimate a parameterized posterior probability distribution, typically modeled as a Gaussian:

$$q_\phi(z|x) \approx p_\theta(z|x) \tag{1}$$

where $p_\theta(.)$ represents the true posterior distribution and $q_\phi(.)$ denotes its approximation learned by the encoder. The decoder then samples from this learned posterior distribution and processes the sample to reconstruct the input. Formally, the decoder learns the likelihood distribution $p_\theta(x|z)$ enabling the generation of new samples that closely resemble the training data.

The VAE framework has inspired numerous architectural variants, including conditional VAE, hierarchical VAE, beta-VAE, adversarial autoencoders, VAE with normalizing flows, importance weighted autoencoders, VAE with structured latent spaces, and VAE with disentangled representations, each tailored to address specific challenges in generative modeling.

B. Generative adversarial network

Introduced by Goodfellow et al,[26] GANs implement a distinctive competitive learning paradigm through 2 neural networks: a generator (G) and a discriminator (D). The fundamental objective of GANs is to learn data distributions through adversarial training dynamics.

The generator takes random noise as input and transforms it into synthetic data, whereas the discriminator analyzes both authentic training data and the generator's synthetic outputs, attempting to differentiate between real and generated samples. Simultaneously, the generator aims to produce increasingly convincing outputs that the discriminator cannot distinguish from authentic data. This adversarial process continues iteratively until convergence is achieved (Fig. 2B).

Despite their effectiveness, GANs face significant challenges, particularly mode collapse—a phenomenon where the generator fails to capture the full diversity of the data distribution, leading to limited variety in generated samples with certain distribution modes being underrepresented or entirely absent. Various solutions have been proposed to address this limitation, including minibatch discrimination, conditional GANs, diversity-sensitive loss functions, architectural modifications, regularization techniques, and auxiliary task incorporation.

The GAN architecture has evolved into multiple variants, including Wasserstein GAN,[27] Deep Convolutional GAN,[28] adversarial autoencoder models,[25] and long short-term memory GAN, each addressing specific aspects of the generative challenge.

1. Generative transformer

The generative transformer represents a hybrid neural network architecture that integrates elements from both transformer models and GANs. Generative pretraining transformers (GPTs) have transformed natural language processing (NLP) and significantly influenced broader AI applications through their sequence-to-sequence capabilities in tasks such as text summarization, generation, and translation.[29]

Based on the transformer architecture introduced by Vaswani et al,[29] these models employ self-attention mechanisms to model relationships between different sequence elements, effectively capturing long-range dependencies and contextual information. The distinguishing feature of GPTs is their pretraining methodology—before task-specific fine-tuning, these models undergo training on massive text corpora to develop generalizable features applicable across diverse NLP tasks, creating a rich foundation of transferable knowledge.

C. Diffusion models

Diffusion models constitute a class of generative approaches designed to create new data by reversing a diffusion process.[30] The core principle involves gradually corrupting original samples by adding noise, then learning to reverse this process by denoising corrupted samples to generate new instances.[31] These models are categorized into 3 frameworks: (1) DFPMs, (2) noise-conditioned score networks, and (3) score stochastic differential equations.[31]

1. Denoising diffusion probabilistic models

DFPMs, introduced by Blei et al,[32] implement a dual Markov chain approach. The forward chain progressively adds Gaussian noise to the original sample until it resembles pure white noise, whereas the reverse chain learns to systematically remove noise to recover the original sample (Fig. 2C).

If we denote the original sample as x_0 with distribution $q(x_0)$, the forward Markov chain's transition probability is defined as:

$$q(x_t|x_{t-1}) = N\left(\sqrt{1-\beta_t}\,x_{t-1}, \beta_t \mathrm{I}\right) \tag{2}$$

where $N(\mu, \Sigma)$ represents a normal distribution with mean vector μ and covariance matrix Σ. The parameter β_t indicates the variance added to the sample at time step tt. Notably, this variance can vary between steps, necessitating a carefully designed variance schedule.

The distribution of the sample at time step tt, conditioned on the original sample, is computed as:

$$(x_t|x_0) = N(\sqrt{\overline{\alpha}_t}\,x_0, (1-\overline{\alpha})\mathrm{I}) \tag{3}$$

w here $\overline{\alpha}_t$ is defined as:

$$\overline{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s) \tag{4}$$

This formulation enables the systematic corruption and reconstruction of molecular structures, providing a powerful framework for generative drug design.

IV. Linear representation

This section examines generative approaches that employ linear representation as the foundational representation method for drug molecules.

A. Encoder-decoder

Gómez-Bombarelli et al[33] pioneered a method representing drugs as SMILES strings fed into a VAE. In this framework, the encoder processes SMILES input and maps it to a hidden representation, which the decoder subsequently transforms back into the original molecular space. Notably, this hidden representation is also utilized by an auxiliary network to develop a more discriminative model for molecular property prediction.

Kusner et al[34] introduced grammar VAE specifically for generating drugs in SMILES representation. Their approach maps SMILES strings to 1-hot representations using SMILES grammar for encoder input. The decoder output is then mapped to a sequence of production rules, generating molecules in SMILES format.

Addressing the nonunique nature of SMILES strings, Alperstein et al[35] developed a method that encodes multiple SMILES representations of a single molecule into a fixed-length latent representation. During decoding, this representation generates a disjoint set of valid SMILES strings for the molecule.

Mohammadi et al[36] created a Penalized VAE for drug design using SMILES representation, incorporating a decoder term that penalizes imbalances in the character distribution of SMILES strings.

Leveraging research showing that chemical compounds impact gene transcription, Shayakhmetov et al[37] developed an adversarial autoencoder model generating drug molecules that induce specific gene expression changes. Their approach separates gene expression features into related and unrelated sets, using the related features to generate drug-like molecules with SMILES as both input and output representations.

Pravalphruekul et al[38] introduced BiCEV, a de novo drug design approach employing 2 VAEs—one processing SMILES input and another processing gene expression profiles. After the encoders learn their respective distributions, both learned distributions are fed into the decoders.

Putin et al[39] presented the Reinforced Adversarial Neural Computer for de novo small molecule design, combining GANs with reinforcement learning (RL). Reinforced Adversarial Neural Computer employs a differentiable neural computer as its generator, leveraging its explicit memory capabilities to enhance the generation process.

Zapata et al[40] developed a de novo drug design model that first trains a VAE to reconstruct drugs in SELFIES format. Subsequently, they train a GAN processing noise alongside cell painting images (morphological profiles), enabling the generator to learn a distribution over molecular embeddings conforming to morphological patterns. The learned embedding vector is then fed into the decoder to generate the final molecule.

Other researchers have focused on utilizing gene expression profiles to generate novel drug-like molecules.[41,42] Bagal et al[43] introduced LigGPT, employing a transformer-decoder model for molecular generation, with both input and output represented in SMILES format. The model is trained on next-token prediction tasks to generate drug-like molecules.

Mao et al[44] developed a dual-transformer architecture for molecule generation: the first transformer maps International Union of Pure and Applied Chemistry molecular representations to a distribution, whereas the second transforms this representation into SMILES format. This approach effectively explores the relationship between International Union of Pure and Applied Chemistry nomenclature and SMILES representation, demonstrating particular efficacy in antiviral drug discovery.

Wang et al[45] created a GPT-based model representing molecules in SMILES format, pretrained on molecular sets (MOSES) and Research Collaboratory for Structural Bioinformatics Protein Data Bank datasets. Building on this concept, Li et al[46] introduced DrugGPT, adapting the GPT-2 model with a redesigned tokenizer. Their approach demonstrates stable learning characteristics while avoiding mode collapse issues, resulting in strong generalization capabilities.

Hu et al[47] developed MolRL-MGPT, a multiagent drug design system where each agent functions as a GPT model processing SMILES representations with position embeddings to generate SMILES sequences. The agents collaborate to identify optimal molecular candidates.

Kaitoh and Yamanishi[48] created a drug design method incorporating initial source molecule selection, feeding these molecules into a VAE to learn generative distributions. Their source selection module identifies chemically induced profiles with high correlation to target perturbation profiles, enabling more effective molecule generation for specific target proteins.

Wei et al[49] introduced a generative drug design approach based on a blank-filling language model utilizing a generative molecular transformer.

B. Generative adversarial network

A persistent challenge in training generative models on SMILES representations is the low validity rate stemming from SMILES syntax complexity. O'Boyle and Dalke[50] addressed this by introducing DeepSMILES, which eliminates mismatched parentheses by exclusively using closed parentheses, with their quantity indicating branch length. Additionally, DeepSMILES resolves ring closure symbol pairing issues by employing single symbols at ring closure positions, with these symbols denoting ring size.

Building on this innovation, Xu et al[51] developed DeepGAN, utilizing DeepSMILES rather than conventional SMILES representation. Their method combines GAN architecture with RL, employing strategy gradient optimization for rewards and adversarial loss.

Abbasi et al[52] implemented a process where SMILES input feeds into an encoder-decoder to generate hidden representations. They then trained a WGAN-GP network where the generator learns representations, whereas the discriminator classifies encoder-decoder hidden representations as real samples and generated samples as fake. The generated samples subsequently feed into the decoder to produce molecules, with a feedback loop updating the network and selecting Pareto optimal molecules using the nondominated sorting genetic algorithm-II.

Lee et al[53] introduced SMILES-MaskGAN, generating adversarial networks through RL. During training, parts of input SMILES sequences are masked, then fed to the generator to learn probability distributions over masked positions conditioned on visible segments, enabling complete molecule creation.

Prykhodko et al[54] developed LatentGAN, which processes SMILES input and integrates an encoder-decoder architecture with GAN functionality. During testing, samples from the GAN generator feed into the heteroencoder's decoder to generate de novo molecules.

C. Diffusion-based model

Peng and Zhu[55] pioneered DIFFUMOL, a groundbreaking diffusion-based model representing drugs in SMILES format. This innovation merges transformer architecture with diffusion

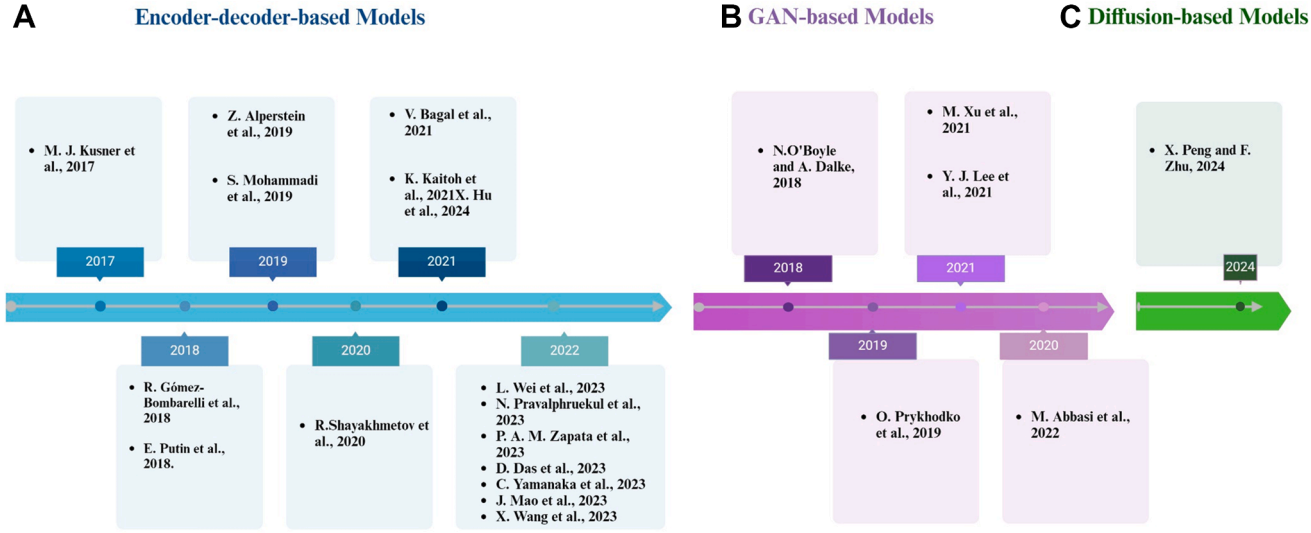


Fig. 3. The distribution of approaches that use a drug's linear representation over time. Approaches utilize the (A) encoder-decoder-based models, (B) GAN-based models, and (C) diffusion-based models to generate a new drug molecule.

models, adding noise to molecule feature vectors (derived from the transformer) during the diffusion process's forward pass. The integration of diffusion modeling with transformer architecture facilitates fine-grained molecular generation with significant applications in drug discovery and molecular modeling.

D. Discussion

This section has examined various generative methods utilizing linear representation for drug molecule generation. As illustrated in Fig. 3, diffusion-based models are notably underrepresented in this domain. This limited adoption can be attributed to diffusion models' optimization through progressive noise addition and denoising—a process poorly suited to SMILES sequences due to their structured syntax and rule-based nature.

A critical challenge in this field is ensuring that generated SMILES sequences correspond to valid, rule-compliant, interpretable molecules. Addressing this issue, Xu et al[51] introduced UnCorrupt SMILES, a postprocessing correction method for generated SMILES. Their evaluation across 4 generative models—recurrent neural network (RNN), target-directed RNN, GAN, and VAE—demonstrated significant improvements in model validity.

Figure 3 reveals the predominance of VAE-based approaches compared with GAN-based methods, attributed to GANs' lower validity rates and limited molecular diversity. Analysis of GAN-based methods shows that successful implementations typically incorporate auxiliary models designed to mitigate mode collapse (thus enhancing variety) while maintaining high validity rates.

Linear molecular representations like SMILES and SELFIES present unique challenges and opportunities for generative models. Among the 3 main architectures, VAEs have emerged as the most suitable for linear representations due to their ability to learn continuous latent spaces from sequential data while maintaining reasonable validity rates (70%–90%). VAEs leverage recurrent or transformer-based networks to encode and decode SMILES strings, enabling scaffold hopping and property optimization through latent space interpolation. Their training stability and controllability make them a robust choice, although they can struggle with diversity. GANs while capable of producing high-quality, diverse molecules, often suffer from mode collapse and low validity rates (50%–80%) unless reinforced with techniques like gradient penalty or RL. Their training instability and need for careful tuning limit broader applicability. Diffusion models, although promising for their ability to generate high-diversity outputs, are less naturally suited to discrete token sequences like SMILES, leading to high invalidity rates (50%–70%) and slow generation times. Recent hybrid approaches, such as diffusion transformers, aim to mitigate these issues but remain experimental. For most applications, VAEs strike the best balance, whereas GANs and diffusion models are reserved for specialized cases requiring extreme diversity or novel exploration of chemical space. Future advancements may focus on hybrid architectures or improved tokenization (eg, SELFIES) to enhance performance across all models.

As summarized in Table 1, the choice of generative architecture for linear representations involves fundamental compromises. VAEs offer the most robust controllability for medicinal chemistry tasks, whereas GANs—despite their instability—unlock broader chemical diversity. Diffusion models, although computationally intensive, represent a paradigm shift for de novo exploration, particularly when combined with 3D constraints.

Table 1
Comparative performance of generative architectures for linear molecular representations (SMILES/SELFIES)
Key metrics include diversity, training stability, controllability, generation speed, syntax awareness, and optimal use cases for VAEs, GANs, and diffusion models.

Metric	VAEs	GANs	Diffusion Models
Diversity	Moderate	High (when stable)	Very high
Training stability	Stable	Unstable (requires tuning)	Computationally intensive
Controllability	High (latent space ops)	Moderate (requires RL)	Emerging (promising)
Generation speed	Fast (single pass)	Fast	Slow (iterative)
Syntax awareness	Moderate (improved with SELFIES)	Low (needs postprocessing)	Moderate (better with hybrid)
Best use case	Scaffold hopping, optimization	High-diversity generation	Novel exploration, 3D hybrid

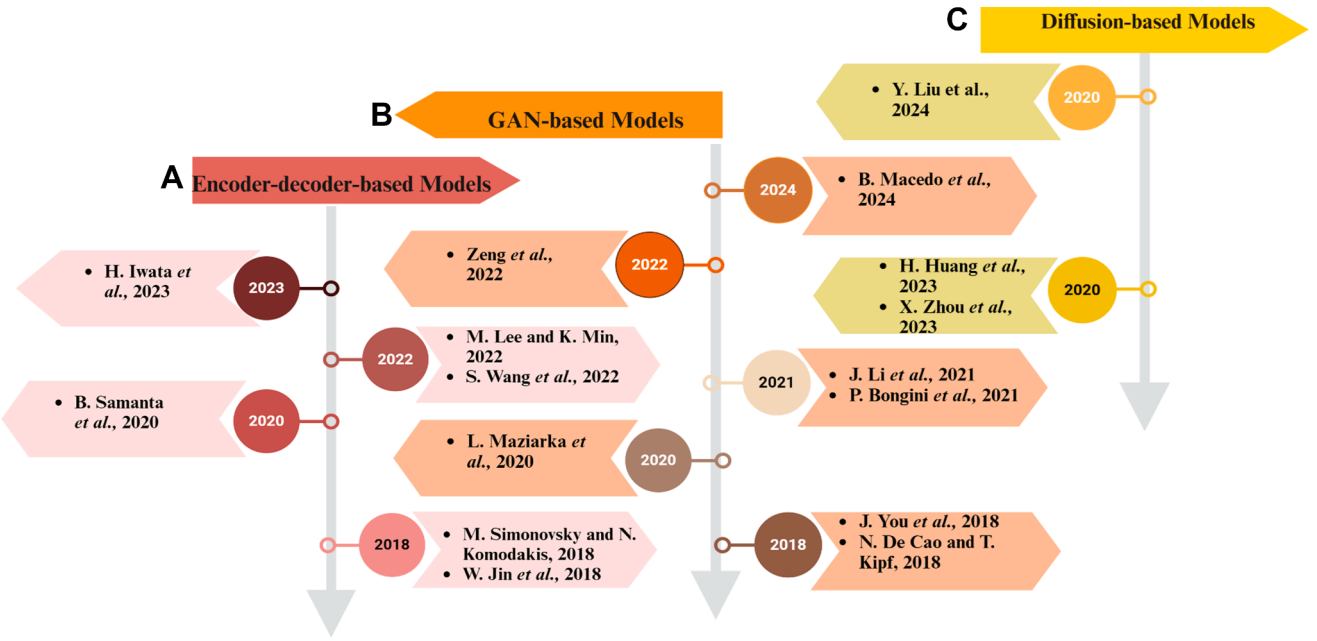


Fig. 4. The distribution of the approaches over time that use the 2D representation of a drug. Approaches utilize the (A) encoder-decoder-based models, (B) GAN-based models, and (C) diffusion-based models to generate a new drug molecule.

V. 2-Dimensional graph representation

The representation of molecules as 2D graphs offers significant advantages in generative drug design, enabling more comprehensive encoding of molecular properties compared with linear representations. This section examines generative approaches leveraging a 2D graph representation for drug molecule generation.

A. Autoencoder

Simonovsky and Komodakis[56] pioneered GraphVAE, applying a VAE architecture to generate molecules in a 2D graph representation. Their approach represents molecules using 3 matrices: node feature matrix, edge feature matrix, and adjacency matrix. The encoder maps this input into a continuous hidden representation, from which the decoder samples to generate a probabilistic, fully connected graph.

Samanta et al[57] introduced neural embedding VAE (NEVAE), a deep generative model for molecular graphs that represents molecules as 2D graphs with weighted edges. Their decoder generates molecules with spatial coordinates of atoms, and they developed an optimization algorithm for generating molecules with desired properties by optimizing the decoder.

Jin et al[58] developed a more structured approach, first extracting a junction tree from a 2D graph input. This junction tree feeds into a VAE where the encoder learns a hidden embedding space and the decoder reconstructs the junction tree. Simultaneously, they feed the 2D graph into the encoder to learn a molecule embedding space. The final reconstruction combines the reconstructed junction tree with the molecule embedding space to generate a complete molecular 2D graph.

Lee and Min[59] created the molecular graph conditional VAE, which processes an initial 2D molecular graph representation (node feature matrix and adjacency matrix) alongside a condition vector through a VAE. Their method feeds both the hidden embedding space and the condition vector into the decoder to generate 2D molecules with specific properties.

Iwata et al[60] developed variational graph autoencoder (VGAE)-Monte Carlo tree search (MCTS), combining a VGAE with MCTS. Their model takes a 2D molecular graph as input and uses VGAE to generate 2D representations, including node and edge information matrices. These matrices then feed into a module employing MCTS to develop bonds between atoms sequentially.

Wang et al[61] introduced the molecular substructure tree generative model, representing molecules as substructure trees that incorporate chemical specificity. This approach generates molecules iteratively by adding substructures in each iteration, allowing for more controlled synthesis of complex molecular structures.

B. Generative adversarial network

You et al[62] developed the Graph Convolutional Policy Network for generating molecules with desired properties, implementing a RL-based model with a policy gradient training strategy. This approach leverages Graph Convolutional Networks to process and generate molecular graphs.

De Cao and Kipf[63] introduced MolGAN, a GAN-based model for drug molecule generation. The generator produces a dense adjacency tensor and an annotation matrix (node feature matrix) representing the molecular structure. They incorporated a reward network to predict rewards for generated molecules, guiding the system toward creating drug molecules with desired properties.

Macedo et al[64] presented MedGAN, a de novo drug design method utilizing Wasserstein GAN to generate new quinoline scaffold molecules. Their approach represents molecules as 2D graphs and implements Graph Convolutional Networks within the Wasserstein GAN architecture to enhance molecular generation.

Maziarka et al[65] developed Mol-CycleGAN, which takes 2D graphs as input and, given a starting molecule, generates structurally similar molecules with desired characteristics conditioned on the input molecule. Their research demonstrates that graph-based representation significantly improves the generation of valid compounds compared to SMILES representation.

Table 2
Comparative performance of generative architectures for 2D graph representations
Key metrics include topology control, fragment awareness, 3D consistency, training stability, generation speed, and best use cases for VAEs, GANs, and diffusion models.

Metric	VAEs	GANs	Diffusion Models
Topology control	Strong (hierarchical generation)	Moderate (struggles with cycles)	Excellent (preserves connectivity)
Fragment awareness	High (scaffold-based)	Low (global generation)	High (local edits)
3D consistency	Weak (pure 2D)	Weak (pure 2D)	Moderate (3D-aware variants)
Training stability	Stable	Unstable (mode collapse)	Moderate (memory-intensive)
Generation speed	Fast (single pass)	Fast	Slow (100–1000 steps)
Best for	Scaffold hopping, lead optimization	Diverse library generation	High-fidelity target-aware design

Zeng et al[66] created ImageMol for molecule property prediction, leveraging large-scale molecular images and chemical knowledge to extract molecular structures. Their approach employs 5 pretraining strategies for the molecular encoder: (1) a structural classifier for chemical structural information prediction, (2) a rationality classifier differentiating rational from irrational molecules, (3) a jigsaw classifier for permutation consideration, (4) mask-based contrastive learning to increase similarity between masked and original images, and (5) a generator mapping latent features to molecular images. This pretrained encoder subsequently fine-tunes for task-specific features.

Li et al[67] implemented a qubit-efficient quantum GAN with a hybrid generator to generate drug molecules in 2D representation using atom vectors and bond matrices, demonstrating the potential of quantum computing in molecular design.

Bongini et al[68] developed a sequential molecular graph generator that processes partial 2D graphs as input and progressively adds nodes or node groups with their corresponding edges to build complete molecular structures.

C. Diffusion-based models

Huang et al[69] pioneered the application of conditioned diffusion models for learning discrete graph structures to generate new drug molecule graphs. Their approach represents molecules using a node features matrix and a 2-channel edge matrix (encoding edge type and edge existence). The model defines a forward diffusion process through stochastic differential equations on both feature and edge-type matrices, with the inverse generative process employing discrete graph structure conditioning.

Liu et al[70] introduced graph diffusion policy optimization, a diffusion process for graphs capable of handling arbitrary differentiable or nondifferentiable objectives through RL. Their evaluation of graph diffusion policy optimization for de novo drug design demonstrated performance comparable with state-of-the-art models.

Zhou et al[71] developed DecompOpt, an iterative 4-step process: (1) sampling a reference arm from an ordered arm list conditioned on subpocket, (2) generating ligand molecules based on protein subpocket and the selected arm using decomposable diffusion, (3) decomposing generated ligand molecules into scaffolds and arms, and (4) updating the ordered arm list. DecompOpt represents input as a 2D graph using node and edge feature matrices, with the edge feature matrix implemented as a 3D structure where each edge has an associated feature vector.

D. Discussion

Recent research confirms that graph-based molecular representation provides superior expressiveness and preserves more molecular properties compared with linear SMILES representation.[16] As shown in Fig. 4, which illustrates the chronological development of approaches utilizing 2D graph representation, GAN-based models predominate in 2D molecular graph generation.

A significant challenge in this domain involves the proper representation and handling of the edge matrix. In some methodologies, the edge matrix simply corresponds to the graph's adjacency matrix, providing information about edge presence. However, edge type and feature information also contain valuable knowledge, and several approaches have attempted to incorporate this information to enhance molecular generation.

2D graph representations offer several key advantages over linear notations for molecular generation while presenting unique challenges. By directly encoding atoms as nodes and bonds as edges, graph-based approaches naturally preserve chemical validity without requiring the complex syntax rules needed for SMILES or SELFIES generation. This inherent structural fidelity typically yields higher validity rates (often exceeding 90%) compared with linear methods. The graph format also enables more chemically intuitive generation strategies, such as scaffold-based construction and fragment linking, which better mimic real-world drug design processes. However, 2D representations come with important limitations—they lack explicit 3D spatial information, which can result in generated structures with unrealistic geometries or steric clashes. The discrete, combinatorial nature of graph generation also introduces computational complexity, often requiring sophisticated autoregressive or RL approaches to manage the sequential addition of atoms and bonds. Although graph neural networks provide powerful tools for processing these representations, they demand greater memory and processing resources compared with sequence-based models, particularly for larger molecules. Among generative architectures, VAEs have proven particularly effective for graph generation due to their ability to learn continuous latent spaces of valid substructures, although they may produce less diverse outputs. Diffusion models are emerging as promising alternatives through graph-structured denoising approaches, whereas GANs remain less common due to difficulties maintaining both graph connectivity and chemical validity during adversarial training. These characteristics make 2D graph representations particularly well-suited for structure-based drug design applications where chemical validity and interpretable generation processes are prioritized.

The comparative performance of generative architectures for 2D molecular graphs is systematically evaluated in Table 2, which highlights 4 key findings. First, VAEs demonstrate fragment awareness due to their hierarchical generation approach, making them ideal for scaffold hopping applications (see Table 2, Fragment Awareness column). Second, although GANs achieve faster generation speeds, their inherent instability and moderate topology control limit their standalone utility without RL augmentation. Third, diffusion models emerge as the most robust option for high-fidelity generation through iterative refinement, although at a significant computational cost. Finally, all architectures share a common limitation in 3D spatial consistency when using pure 2D representations, as noted in the 3D consistency row.

Figure 4 demonstrates that all 3 generative method categories—encoder-decoder, GAN-based, and diffusion-based—are frequently employed for 2D representation of drug molecules, suggesting the versatility and effectiveness of 2D graph representation across various generative frameworks.

VI. 3-Dimensional graph representation

3D molecular representation provides the most comprehensive structural information, capturing spatial relationships critical for understanding drug-target interactions. This section examines generative approaches leveraging 3D graph representation for drug molecule design.

A. Autoencoder

Ochiai et al[72] developed the natural product-oriented VAE specifically for 3D molecular structures. Their innovative approach decomposes compound structures into fragments according to specific rules, organizing these fragments into tree structures. Both the derived tree structure and the original compound structure feed into the encoder to learn comprehensive molecular distributions.

Liu et al[73] introduced graph multiview pretraining (GraphMVP), a novel approach combining 2D and 3D molecular information. Their methodology employs 2 complementary techniques: first, a contrastive self-supervised learning approach creating 2D/3D pairs belonging to either the same or different molecules, using contrastive loss functions to learn enhanced feature representations; second, generative SSL techniques leveraging global information by generating 3D conformers from 2D molecules through a VAE-like network.

Ragoza et al[74] developed an encoder-decoder architecture conditioned on target binding pockets. Their system utilizes 2 specialized encoders—an input encoder processing drug-protein complexes and a conditional encoder handling protein binding pockets. The outputs from these encoders feed into a decoder that learns atomic density distributions. The generated structures undergo postprocessing through atom type and coordinate modification, followed by molecular structure refinement.

Li et al[75] created DeepLigBuilder for designing molecules targeting specific binding sites. Their approach integrates 2 key components: the Ligand Neural Network (L-Net) module generating valid 3D molecules, and a MCTS module incorporating target binding site information. L-Net functions as an iterative model with a state encoder and policy network (decoder), whereas MCTS computes reward functions by processing intermediate states through selection, expansion, and simulation operations, with information backpropagated through the network to update Q-values.

Imrie et al[76] developed an encoder-decoder model utilizing 3D molecular information, processing starting structures in graph representation alongside 3D pharmacophore maps in voxel grid representation. These inputs feed into specialized graph neural network (GNN) and convolutional neural network, respectively, with their combined outputs serving as decoder input. The decoding process proceeds through sequential node generation steps, iteratively creating molecular bonds.

Luo et al[77] addressed the challenge of incorporating protein target interaction knowledge into structural drug design. Their model takes 3D protein binding sites as input and generates 3D coordinates for drug molecules as output. Their approach gradually builds drug molecules through 2 primary modules: a context encoder processing 3D binding sites and generating substructures to learn representations via graph neural networks, and a spatial classifier predicting atom types for specific spatial positions.

B. Generative adversarial network

Anand and Huang[78] pioneered a generative model for protein structures using pairwise distances between α-carbon coordinates on protein backbones. Their approach feeds random noise to a generator producing pairwise distance matrices, from which α-carbon coordinate positions are extracted using alternating direction method of multipliers.[79] These structures subsequently feed into Rosetta software for refinement.

Song et al[80] developed a GAN-based approach leveraging transfer learning to incorporate target proteins implicitly, directing attention to specific regions of chemical space. Their methodology represents molecules in 3D using discretized heavy atom positions relative to ligand geometric centers, described through 19 atom features. They implemented the Wasserstein GAN architecture to generate diverse drug-like molecules.

Wang et al[81] introduced RELATION, combining an adversarial autoencoder architecture with GAN frameworks. Their system utilizes 3 specialized 3D encoders: a ligand encoder (private), a ligand-protein encoder (shared), and a protein-ligand complex encoder (private). The resulting feature vectors feed into both a discriminator (distinguishing noise from meaningful features) and a long short-term memory–based decoder, with the system's loss function defined to transfer protein-ligand complex knowledge to latent feature spaces for generation.

C. Diffusion-based technique

Xu et al[82] introduced GeoDiff, a geometric diffusion technique for molecule conformation generation. Their approach encodes 3D molecules as undirected graphs and modifies the denoising diffusion process to achieve equivariant reverse generation. They demonstrate that with SE(3)-invariant initial distributions and SE(3)-equivariant Markov transitions, the resulting distributions maintain SE(3)-invariance.

Guan et al[83] developed DecompDiff for 3D molecule design, first decomposing ligand molecules into arms and scaffolds, then modifying diffusion processes to accommodate decomposed priors. Notably, DecompDiff considers both nodes and bonds during diffusion, enabling more comprehensive molecular generation.

Schneuing et al[84] created DiffSBDD, an SE(3)-equivariant diffusion model conditioned on protein pockets for 3D ligand generation. Building on this concept, Torge et al[85] investigated scaffold hopping—creating new molecules by preserving functional groups while exchanging scaffold atoms—using diffusion processes to construct novel scaffolds given protein and molecular functional groups.

Igashov et al[86] introduced DiffLinker, a fragment-based drug design approach utilizing E(3)-equivariant 3D-conditional diffusion models. Their work focuses on effectively linking arbitrary numbers of fragments to create novel molecular structures.

Liu et al[87] developed MolNet-3D, a representation framework for molecules based on 3D topography. Their deep learning network maps 3D topography descriptors to quantum mechanical molecular observables, preserving 3D topographic information while maintaining invariant representation properties.

Corso et al[88] introduced DiffDock, a diffusion-generative model operating over translational, rotational, and torsional degrees of freedom to predict 3D binding structures of ligands with target proteins.

Addressing limitations in existing methods that typically consider only point cloud representations while ignoring interatomic relationships, Huang et al[89] developed the molecular diffusion model (MDM) for 3D molecule generation. Their

Table 3
Performance comparison of generative architectures for 3D molecular design

Metric	VAEs	GANs	Diffusion Models
Conformer sampling	Limited	Moderate	Extensive
Target awareness	Weak	Moderate	Strong
Generation speed	Moderate	Fast	Slow
Best application	Scaffold optimization	Decoy generation	De novo design

approach encodes interatomic relationships through 2 mechanisms: covalently bonded forces (defined when interatomic distances fall below specific thresholds) and van der Waals–bounded forces (defined for atom pairs with spacing above certain thresholds). To model these distinct bond types, they implemented 2 different equivariant networks and introduced a novel latent variable controlling representation during diffusion.

Qiang et al[90] observed that existing methods often neglect intrinsic local structures while focusing exclusively on atom types and features, resulting in poor-quality molecular generation. To address this issue, they implemented a fragment-based generation technique with a coarse-to-fine strategy addressing combinatorial optimization challenges. Their 3-step method first generates coarse-grained molecules through equivariant diffusion, then obtains corresponding fragments for each coarse node via message-passing algorithms, and finally fuses coarse-grained molecules with their corresponding fragments to create complete molecular structures.

D. Discussion

The 3D representation of molecules provides significantly richer structural information compared with 1D and 2D representations. Analysis of temporal distribution patterns for 3D molecular generation approaches reveals that diffusion models have gained predominance over GAN and encoder-decoder architectures in this domain. This trend suggests that in 3D representation, the forward diffusion process of gradually perturbing input through noise addition proves more intuitive and effective. Additionally, diffusion-based models demonstrate superior generation quality with enhanced mode coverage and diversity, making them particularly well-suited for 3D molecular generation tasks.

3D molecular representations offer significant advantages for drug design by explicitly modeling spatial arrangements critical for binding affinity and pharmacokinetics. Unlike 2D approaches, 3D generation captures steric interactions, conformational flexibility, and physiologically relevant geometries (Table 3). Diffusion models have emerged as the leading architecture for this task, achieving 90%–98% structural validity through SE(3)-equivariant networks that preserve molecular symmetry during iterative denoising. Although slower (requiring 100–500 refinement steps), these models excel at target-aware generation and multiconformer sampling. VAEs provide faster (single pass) generation but struggle with geometric accuracy and often produce unrealistic clashes. GANs, although capable of rapid decoy generation, remain limited by instability and require extensive RL refinement. All 3D methods face shared challenges, including computational intensity (10–100× slower than 2D) and dependence on scarce experimental structural data.

Table 3 highlights critical tradeoffs: diffusion models provide the most physically accurate structures but require substantial computational resources, whereas VAEs offer practical efficiency for less geometry-sensitive tasks. Emerging solutions combine these approaches—for instance, using VAEs for initial scaffold generation followed by diffusion-based refinement—to balance speed and accuracy. Future advancements will likely focus on integrating quantum mechanical constraints and expanding training datasets to address current limitations in conformational diversity and synthetic accessibility.

VII. Scalability

The scalability and computational demands of molecular generation methods are critical considerations for their practical application, particularly in large-scale drug discovery. Template-based approaches, which ensure synthetic feasibility through predefined reaction templates and building blocks, vary significantly in their efficiency and scalability.

Gaiński et al[91] investigate scalable and cost-efficient de novo template-based molecular generation. RGFN[92] relies on a small, curated fragment library, which limits its chemical diversity but ensures low computational overhead by avoiding expensive retrosynthesis validation. In contrast, SynFlowNet[93] struggles with scalability when applied to large building block libraries due to its reliance on RL for trajectory validation, which introduces significant computational costs. RxnFlow[94] performs better than SynFlowNet but still lacks efficiency in large library settings, as its uniform exploration strategy does not incorporate cost or synthetic feasibility guidance. SCENT[91] has been designed to address these limitations. By employing recursive cost guidance, the need for expensive recursive evaluations is reduced through the use of auxiliary machine learning models. The dynamic library mechanism further enhances scalability by reusing high-reward intermediates, thereby amortizing the cost of synthesis tree construction. Additionally, the exploitation penalty ensures balanced exploration without the computational burden of pseudo-counting methods. As a result, SCENT efficiently scales to large libraries (tested up to 128k building blocks) while maintaining low computational overhead. Unconstrained generative models, such as REINVENT[95] and Graph GA,[96] offer high theoretical scalability but suffer from practical limitations. Although they can generate diverse molecules, a significant portion of their outputs are often unsynthesizable, necessitating costly post hoc validation. REINVENT, for instance, requires retrosynthesis checks (eg, the AiZynthFinder software[97]), which drastically increase computational demands. Graph GA, although computationally lightweight during generation, yields a low percentage of synthesizable molecules, making it impractical for real-world applications. In summary, based on the investigation of Gaiński et al,[91] SCENT provides an optimal balance between scalability, computational efficiency, and synthetic feasibility. Its guided exploration

Table 4
Performance-resource tradeoffs in MPNN++, transformer, and hybrid GPS++ graph neural networks

Aspect	MPNN++	Transformer	GPS++
Compute/GPU	Low (sparse ops)	High (dense attention)	Moderate
Memory	Low	Very high	Medium
Depth scalability	Limited (~16 layers)	High (>32 layers)	High (hybrid)
Data efficiency	High	Low (needs more data)	Moderate
Best use case	Small graphs, low budget	Global dependencies	Foundational models

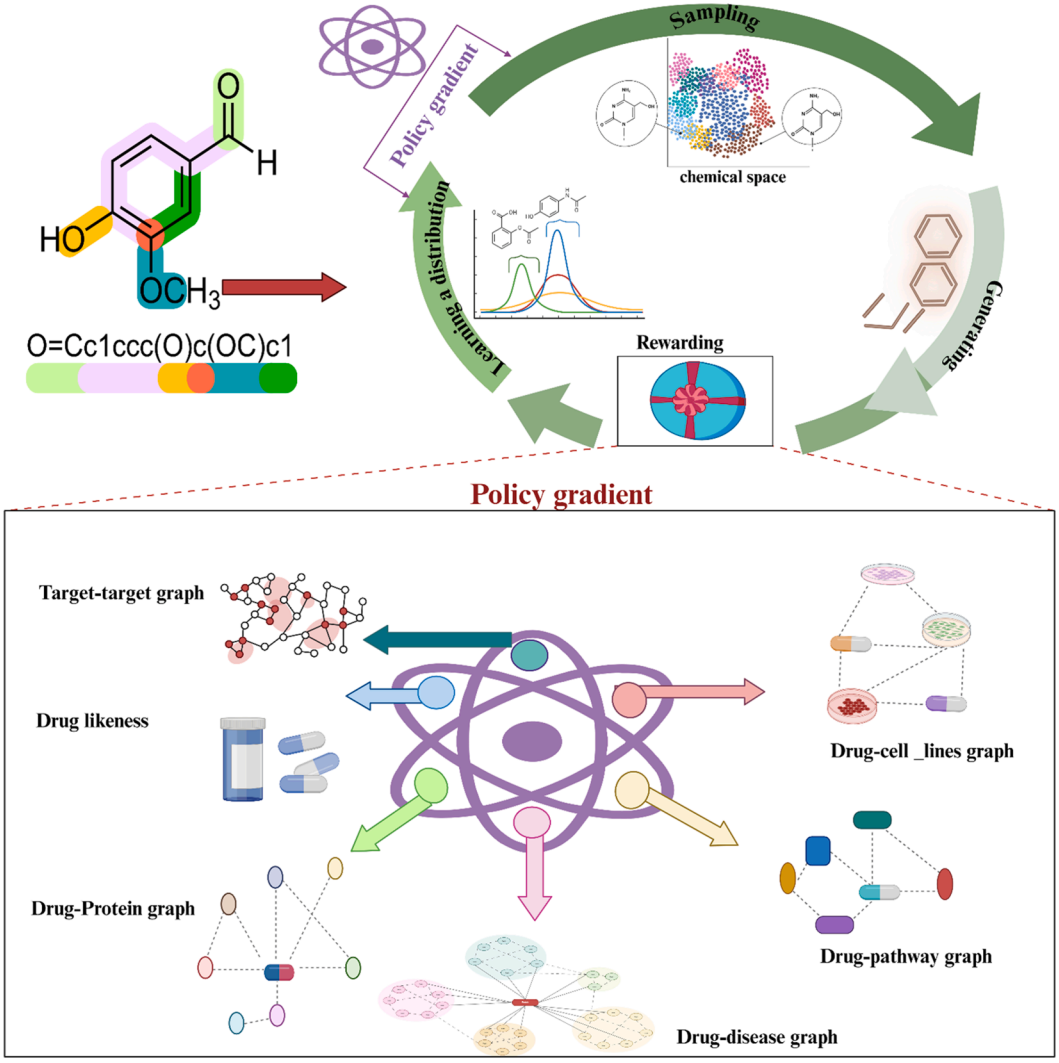


Fig. 5. The impact of reinforcement algorithm in generating molecule with desires properties.

and dynamic library mechanisms make it well-suited for large-scale molecular generation, outperforming both constrained and unconstrained baselines in practical settings.

Sypetkowski et al[98] investigate the computational requirements and scalability considerations for the different GNN architectures—message passing neural network (MPNN)++, Transformers,[99] and the hybrid graph property synthesis (GPS)++.[100] They have shown that they are significantly influenced by their fundamental design differences. MPNN++ models are characterized by greater parameter efficiency and lower computational demands, as localized message-passing operations are utilized, which scale linearly with graph size. However, limitations in depth scalability are encountered due to over-smoothing, with performance typically plateauing around 16 layers (Table 4).

In contrast, graph transformers exhibit higher computational costs because of their global self-attention mechanisms, which scale quadratically with the number of nodes. This results in substantial memory requirements, particularly for large graphs, although better depth scalability is achieved through residual connections and reduced over-smoothing. The hybrid GPS++ architecture is designed to balance efficiency and expressivity, combining the localized feature extraction of MPNNs with the global context modeling of Transformers, making it well-suited for large-scale pretraining while requiring careful optimization of the MPNN-to-transformer ratio.

Training these models at scale, especially for configurations with 1B and 3B parameters, necessitates significant computational resources. For example, pretraining a 1B-parameter MPNN++ on the LargeMix dataset (5M molecules) is conducted using distributed training across multiple graphics processing units (GPUs) (eg, 8 NVIDIA A100s) with gradient accumulation to maintain effective batch sizes. The 3B models further require advanced parallelism strategies, such as tensor or pipeline parallelism, to address GPU memory constraints. Pretraining durations can extend to several weeks, highlighting the importance of optimizations like mixed-precision training and gradient checkpointing.

Fine-tuning and probing, although less resource-intensive, still benefit from scalable implementations. Probing, in particular, is computationally lightweight due to its frozen backbone and simple multilayer perceptron head. Key scalability insights indicate that width scaling (increasing model parameters) consistently enhances performance but is quickly limited by GPU memory, whereas depth scaling is more architecture-dependent. MPNNs demonstrate diminishing returns beyond a certain depth, whereas transformers and GPS++ models scale more effectively with additional layers.

Data scaling is observed to yield sublinear performance improvements, with the 5M-molecule LargeMix dataset proving sufficient for 1B models but potentially requiring expansion (eg, via Phenomics data) for 3B configurations. Multitask learning, involving thousands of labels, introduces additional computational overhead but is essential for achieving robust downstream performance.

For practical deployment, MPNN++ is recommended for resource-constrained settings or small graphs, Transformers for tasks requiring global context (given sufficient computational resources), and GPS++ as a balanced choice for foundational models. Future work could explore dynamic sparsity techniques, enhanced parallelism strategies, and hardware-aware optimizations to further improve scalability. These considerations are critical for the effective application of large-scale GNNs in real-world scenarios, particularly in domains such as drug discovery, where the tradeoff between efficiency and performance must be carefully managed.

VIII. Pharmacological aspect

Generative drug design models offer unprecedented capabilities for creating novel molecules with specific pharmacological properties. This section examines how these approaches influence pharmaceutical development through 2 primary mechanisms: property optimization and target identification.

A. Property optimization and target identification

Generative models contribute to drug discovery through 2 main pathways: first, property optimization utilizes these models to generate molecules with specific pharmacological profiles, optimizing parameters including potency, safety, and metabolic stability. Second, target identification employs generative models to identify potential drug targets by designing molecules that interact with specific biological pathways or receptors.[101,102]

For property optimization, generative models can integrate pharmacokinetic and pharmacodynamic data to simulate drug behavior in biological systems, facilitating dose optimization and scheduling. These models also predict absorption, distribution, metabolism, and excretion properties, substantially improving drug candidate selection processes.[103]

RL has emerged as a particularly promising approach for optimizing drug properties within de novo molecular design. In this framework, RL algorithms generate novel molecules that maximize specific reward functions combining desirable properties such as potency, selectivity, solubility, and synthesizability.[103,104] The RL agent navigates the vast chemical space through iterative molecule generation, property evaluation, and policy adjustment to increase the probability of generating high-scoring compounds. By incorporating domain-specific knowledge and constraints, including chemical feasibility and toxicity filters, RL can generate molecules optimized for specific targets while meeting requirements for downstream development and clinical testing (Fig. 5).

B. Multiparameter optimization

Multiparameter optimization (MPO) represents one of the most challenging aspects of drug design, requiring simultaneous optimization across multiple biological activity objectives. Winter et al[105] introduced a multiobjective molecule optimization approach that transforms starting compounds into modified molecules with enhanced properties. They developed a novel objective function combining outputs from multiple prediction models, focusing on molecular substructures.

Gao et al[106] developed the generative network complex, an approach that optimizes multiple chemical properties simultaneously. To demonstrate generative network complex reliability, they generated numerous drug candidates targeting BACE1 inhibitors and produced iterative enhancements of 8 market-approved drugs: ceritinib, ribociclib, acalabrutinib, idelalisib, dabrafenib, macimorelin, enzalutamide, and panobinostat.

Perron et al[107] demonstrated how AI technology, particularly deep learning generative models, accelerates the discovery of lead compounds meeting 11 different biological activity objectives simultaneously. Their AI de novo design algorithm, combined with quantitative structure-activity relationship (QSAR) models for each objective, generated 150 virtual compounds predicted to exhibit activity across all targets. Experimental testing of 11 synthesized molecules showed that AI-designed compounds significantly outperformed initial molecules, achieving an 86% average success rate across all objectives compared with 58% for the initial set.

This study[108] presents a novel RL framework enhanced with active learning to significantly improve the efficiency of MPO in molecular generation. By integrating the REINVENT molecular generator with an active learning system, the approach achieves 5−66× improvements in hit discovery rates while reducing computational time by 4−64× compared with standard RL methods. They have developed key innovations, including a probabilistic MPO acquisition strategy to better balance competing molecular properties (eg, binding affinity vs drug-likeness), optimized batch size ratios between active learning and RL components, and a zero-weight update scheme for surrogate predictions to prevent error propagation. The system demonstrates robust performance across different scoring functions, maintaining molecular diversity while substantially improving MPO scores (1.2−1.87× enhancement). It successfully bridges the gap between AI-generated molecules and practical drug discovery by enabling the use of computationally intensive scoring methods that were previously impractical for RL-based molecular design, representing a significant advance in sample-efficient, multiobjective molecular optimization.

Guo et al[109] present Link-INVENT, an extension of the REINVENT platform that leverages RL for de novo linker design in drug discovery. Built on an RNN-based encoder-decoder architecture, the model generates linkers conditioned on input molecular subunits while optimizing MPO through a customizable scoring function. Key innovations include linker-specific controls for length, flexibility, branching, and rigidity, as well as integration with docking constraints to retain critical protein-ligand interactions. Demonstrating broad applicability, Link-INVENT excels in fragment linking (eg, designing CK2α inhibitors with retained hydrogen bonds), scaffold hopping (eg, optimizing central nervous system-penetrant JNK inhibitors), and PROTAC design (eg, tuning linker properties for Bcl-2/Mcl-1 degraders). Compared with prior methods like DeLinker[110] and SyntaLinker,[111] it offers superior flexibility in RL-driven optimization and generates chemically diverse, high-scoring molecules. The open-source tool bridges deep learning with practical drug discovery, enabling efficient linker design for complex therapeutic targets.

A RL-driven de novo molecular design framework is introduced in the study of Hou[112] that integrates direct preference optimization[113] and curriculum learning[114] to address key challenges in drug discovery, such as training instability, exploration inefficiency, and multiobjective optimization. Traditional RL methods struggle with slow convergence and reward design complexity, but direct preference optimization—adapted from NLP—bypasses explicit reward modeling by optimizing likelihood

Table 5
Molecule generating studies' tasks, datasets, and evaluation metrics

Work	Task	Dataset	Evaluation Metric
DEVELOP[76]	Linker design and scaffold elaboration	ZINC[142] CASF-2016[143] PDBbind Refined set (v. 2019)[144]	Validity Uniqueness Novelty 2D property filters Recovery rate 3D similarity SC_{RDKit} RMSD
VGAE-MCTS[60]	Molecular generation	ChEMBL[145] ZINC	Validity Uniqueness Novelty KL divergence FCD QED penalized LogP
Abbasi et al[52]	Molecular generation	ChEMBL ZINC Biogenic[156]	Validity Uniqueness Novelty KL divergence Internal Diversity External Diversity QED SAscore LogP MW
MSTG[61]	Molecular generation	ZINC250K[147]	Validity Uniqueness Novelty MW LogP SAscore QED NA Penalized logP
DNMG[80]	Molecular generation	ZINC Clean Lead	Validity Unique@1k Unique@10k Novelty Filters Internal diversity FCD SNN Frag Scaff LogP QED SAscore NP MW
DeepTarget[148]	Molecular generation	ChEMBL[149] BindingDB[150]	Affinity and docking scores Validity Uniqueness Novelty MW LogP NP SAscore QED
CRTmaccs[151]	Molecular generation	ChEMBL DrugBank[152] iPPI-DB[153]	Validity Novelty Uniqueness FCD SNN MW LogP NP-likeness SAscore QED QEPPI

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Table 5 (*continued*)

Work	Task	Dataset	Evaluation Metric
TransAntivirus[44]	Molecular generation for antiviral lead compound design	PubChem[154]	Validity Uniqueness Novelty FCD Internal diversity QED LogP MW SAscore NP-likeness QEPPI
Gircha et al[155]	Molecular generation	ChEMBL	Validity MW LogP QED SAscore
Zhang et al[156]	Molecular generation	commercial ChemBridge database ChEMBL ZINC20[157]	Docking score Enrichment factor Diversity Docking pose generation Novelty Diversity
BBAR[158]	Molecular generation	ZINC15	MW TPSA LogP QED Validity Uniqueness Docking score
Dong et al[159]	Molecular generation with properties	ChEMBL[160] ZINC	Validity Uniqueness Novelty Advanced structure check performed by the ChemAxon software SAscore LogP pIC50
Subramanian et al[161]	Molecular generation	ZINC	Validity Uniqueness Novelty KL divergence FCD
cMolGPT[162]	target-specific molecular design	MOSES[163] ExCAPE-DB[164]	Validity Uniqueness Novelty Frag SNN TPSA HBD HBA LogP MW QED SAscore
Tree-Invent[165]	Molecular generation	GuacaMol[96]	Validity Uniqueness Novelty FCD KL divergence
CDGS[69]	Molecular generation	ZINC250k QM9[166]	FCD Validity NSPDK Uniqueness Novelty

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Table 5 (*continued*)

Work	Task	Dataset	Evaluation Metric
DiffHopp[85]	Scaffold hopping	PDBbind	Connectivity Diversity Novelty QED SAscore VINA score
TagMol[167]	Target-specific drug discovery	PDBbind refined 2017	Learning analysis
DiffLinker[86]	Molecular linker design	ZINC CASF-2016 GEOM[168] Pockets Dataset[84]	QED SAscore # Rings Validity Uniqueness Novelty RMSD SC_{RDKit} 2D Filters Recovery rate
Haroon et al[169]	Target-specific drug generation	MOSES GuacaMol ExCAPE-DB	LogP TPSA SAscore QED Validity Unique@10 K Novelty
GeoLDM[170]	3D molecule generation	QM9 GEOM	Stability Validity Uniqueness Conditional generation properties
GRAPHAF[171]	Molecular graph generation	ZINC250K QM9 MOSES	Validity Uniqueness Novelty Reconstruction Penalized logP QED Conditional generation properties
Westermayr et al[172]	Molecular generation for organic electronic application	OE62[173]	Elemental distribution Electronic properties
MDM[89]	3D molecular generation	QM9 GEOM-Drugs	Validity Uniqueness Novelty Stability Conditional generation properties
Levy and Rector-Brooks[174]	Fragment-based molecular generation	ZINC15	Validity QED LogP SAscore
MG2N2[68]	Molecular generation	QM9 ZINC	Validity Uniqueness Novelty MW QED LogP
Mol-Zero-GAN[175]	Molecular generation	PDB[176] ProfKin[177] MOSES	Binding affinity QED SAscore Diversity
Luo et al[77]	3D molecule design and linker prediction	CrossDocked[178]	Binding affinity QED SAscore Diversity Similarity Recovery rate Binding affinity
GMTransformer[49]	Molecular generation	MOSES	Validity Uniqueness Novelty Internal diversity SNN FCN Frag Scaff Filters LogP tPSA QED
QGAN-HG[67]	Molecular generation	QM9	Diversity Validity Uniqueness QED LogP SAscore
DeepLigBuilder[75]	3D molecule generation	ChEMBL	Valildity Diversity MW LogP HBD HBA Rotatable bonds QED NPR1 NPR2 SASA Precision Recall Bond length Bond angles Torsion angles RMSD
RELATION[81]	3D molecular generation	ZINC Clean Lead BindingDB ChEMBL PDB	Validity Uniqueness Internal diversity Novelty FCD Docking score

FCD, Fréchet ChemNet Distance; LogP, lipophilicity; NA, number of atoms; NP, natural product; NSPDK, neighborhood subgraph pairwise distance kernel; MSTG, molecular substructure tree generative model; pIC50, negative logarithm of the half-maximal inhibitory concentration; QEPPI, quantitative estimate of protein–protein interaction targeting drug-likeness; QGAN-HG, quantum generative adversarial network with a hybrid generator; SASA, solvent accessible surface area; Scaff, scaffold similarity; SNN, similarity to the nearest neighbor; TPSA, topological polar surface area.

differences between high- and low-scoring molecules, enhancing stability and efficiency. Curriculum learning further refines the process by progressively increasing task difficulty, from scaffold generation to fine-tuning functional groups. The framework's scalability, faster convergence (6× speedup over baselines), and ability to handle complex MPO tasks (eg, balancing affinity, synthesizability, and drug-likeness) highlight its potential for accelerating drug discovery.

C. Advanced optimization approaches

Ramey et al[115] introduced the optimal drug design concept alongside a hybrid physiologically-based pharmacokinetic machine learning model. Optimal drug design modifies generative drug design models to optimize molecules for drug exposure, safety, and efficacy. Their iterative algorithm first generates drug populations using genetic algorithms, refines each molecule through a VAE, and passes learned features to a neural network that generates pharmacokinetic parameters. A calculated score then informs the genetic algorithm for subsequent generations until convergence.

Wang et al[116] compared deep learning models with traditional machine learning approaches, finding that deep learning models demonstrate marked improvements in physicochemical property prediction but tend to generate molecules with lower scaffold diversity. Canal-Alonso et al[117] investigated generative AI's role in predicting medication side effects, preventing unwanted interactions, predicting drug-target interactions, and generating targeted drug designs.

Overhoff et al[118] developed a 2-step approach where drug-proteome interaction signatures computed by CANDO[119] feed into an autoencoder for dimensionality reduction, followed by new drug compound generation through a conditional VAE with custom target objectives. Bechelli and Delhommelle[120] examined AI's broader role in pharmaceutical drug development processes.

Chenthamarakshan et al[121] introduced controlled generation of molecules, designed to generate drug molecules controlled by target affinity, selectivity, and drug-likeness. Controlled generation of molecules integrates a drug generator with a protein-molecule binding affinity predictor, processing learned embeddings from both components. The system generated molecules for 3 SARS-CoV-2 target proteins, evaluating them on novelty at molecular and chemical scaffold levels, docking free energy, toxicity assessment through in silico screening, and synthetic feasibility through chemical retrosynthesis prediction.

D. Generative protein design

The field of pharmacology is undergoing transformation through generative protein design—computational methods creating novel protein structures with desired functions. This approach enables the creation of targeted therapies and potentially life-saving medications by employing advanced algorithms and machine learning techniques to identify and predict protein structures that interact with specific targets, modulating biological interactions to treat or cure diseases.

Generative protein design has diverse applications in pharmacology: designing proteins with enhanced binding affinities to specific targets, modulating protein-protein interactions, and creating entirely new proteins with novel functions. This technology shows particular promise in cancer treatment, where protein-based therapies can selectively target cancer cells while minimizing impacts on healthy tissue.

Additionally, these approaches address drug delivery challenges by creating proteins that selectively transport drugs to specific tissues or cells, improving efficacy while reducing side effects. This capability is especially relevant for central nervous system diseases, where crossing the blood-brain barrier presents significant challenges.

Despite these advances, substantial challenges remain, particularly regarding protein folding complexity. Proteins form intricate 3D structures, and predicting folding patterns and target interactions remains difficult. Although computational power and machine learning advances are addressing these issues, further research is needed to improve accuracy and efficiency.

Ferruz and Höcker[122] reviewed generative protein design approaches utilizing language models, noting how these models—originally developed for NLP—are being fine-tuned for protein families. They highlight the need for custom transformers in protein generation and demonstrate how conditional tags allow protein language models to generate proteins with specific desired properties.

Various protein generative models employ similar architectures to drug generative models, including VAEs, GANs, and diffusion-based approaches. Karimi et al[123] developed a guided, conditional generative model called semisupervised gcWGAN using Wasserstein GANs. Lee et al[124] introduced ProteinSGM, a protein generative model utilizing score-based generative methods that represents protein structure as images and supports both unconditional and conditional generation, producing native-like, experimentally validated structures with secondary structures consistent with generated backbones.

Multiple protein language models have emerged.[125–128] Madani et al[126] developed Progen, generating random sequences and computing their Rosetta energy scores, demonstrating improved performance over random sequences. Rao et al[129] created a model processing protein sequences alongside multiple sequence alignments to leverage evolutionarily related sequences, implementing row and column attention mechanisms across input sequences. Their modified masked language modeling objective, tailored for diverse protein families, enables transformer-based models to predict protein structure in an unsupervised manner.

Lisanza et al[130] introduced ProteinGenerator, a DFPM simultaneously predicting protein sequences and structures, utilizing RoseTTAfold[131] for structure prediction. Notable advances in protein structure prediction have come from RoseTTAFold2,[132] RoseTTAfold,[131] and AlphaFold2.[133] ProteinGenerator incorporates sequence and structural properties to generate optimized outputs.

Unsal et al[134] developed a novel protein representation utilizing protein language models, evaluating it across 4 key tasks: ontology-based protein function prediction, semantic protein similarity assessment, drug-target binding affinity, and protein-protein binding affinity prediction. Elnaggar et al[135] created ProtTrans by adapting 6 transformer models (Electra, T5, Albert, BERT, Transformer-XL, and XLnet), training on over 390 billion amino acid sequences through collaboration between scientists from Munich, NVIDIA, and Google AI. Recent approaches have also focused on visualization tools to understand attention mechanisms and model reasoning processes.[136,137]

E. Transforming protein design and drug discovery: Collaborative frontiers

AI computing leader NVIDIA has established transformative collaborations in protein design and drug discovery. A notable partnership with biotechnology company Evozyne has produced the Protein Transformer Variational AutoEncoder, built on NVIDIA's BioNeMo framework. This generative AI model, customized with Evozyne's proprietary protein data, facilitates the design of synthetic protein variants exceeding natural enzyme capabilities, accelerating functional protein development for human health and sustainability applications.

NVIDIA's Clara platform further demonstrates their commitment to innovation—a GPU-accelerated drug discovery suite integrating AI, data analytics, simulation, and visualization that provides scientists with cross-disciplinary tools to accelerate drug design and enhance understanding of disease mechanisms.

NVIDIA's collaboration with biotechnology pioneer Genentech extends these capabilities, combining Genentech's lab-in-a-loop iterative framework with generative AI to bridge laboratory-computational divides, potentially revolutionizing medicine through groundbreaking molecular designs.

Similarly, 310.ai is advancing protein design through AI and machine learning, developing computational methods that overcome limitations of traditional experimental approaches. Their transformer-based model Mol.E, optimized for NVIDIA GPUs, represents a paradigm shift by directly designing protein sequences based on intended functions, focusing on relationships between protein function, structure, and sequence to create a novel, efficient design strategy.

Table 6
The list of studies that utilize the ZINC dataset

Method	Number	Preprocessing	Version
DEVELOP	418,000 and 427,000 for linker design and scaffold elaboration	Constructed fragment-molecule pairs using standard transformations from matched-molecular pair analysis[180]	ZINC 15
VGAE-MCTS	249,456	Graph structure representation of a molecule converted into a vector	ZINC 15 is used to evaluate the capability of optimizing the physicochemical properties
MSTG	250,000	The molecule is converted to a substructure tree, considering its chemical specificity	ZINC250K
Zhang et al[156]	1.6 billion molecules for evaluation	-	ZINC20
BBAR	249,456	BRICS decomposition[181] for creating 42,255 building blocks	ZINC15
Dong et al[159]	Around 200 million	-	ZINC20 Used for novelty assessment
Subramanian et al[161]	-	-	ZINC[179]
CDGS	249,455	Kekulized Converted to graph	ZIN250K
DiffLinker	439,410 splits from 250,000 randomly chosen molecules	Generated 3D conformer using RDKit Fragmented Filtered by the number of atoms in the linker and fragments, synthetic accessibility, ring aromaticity, and pan assay interference compounds[182]	ZINC[179]
GRAPHAF	250,000	Kekulized	ZINC250K
Levy and Rector-Brooks[174]	100,000 random molecules	Selected drug-like molecules in stock to purchase (4.7 million) Fragmentation by BRICS Selected molecules with fragments in the top 100 most frequent ones. (2,093,767) Filter by the top 3 most seen atom connection points	ZINC15
MG2N2	249,455	Converted to graph	ZINC[179]
Abbasi et al[52]	108,283	Tokenized the SMILES containing regular tokens and combined tokens	ZINC Biogenic
DNMG	330,000 of 1.9 million	LogP $\leq$ 6 200 Da $\leq$ MW $\leq$ 600 Da	ZINC Clean Lead
RELATION	1,200,000	Molecules containing charged atoms or atoms besides carbon, nitrogen, oxygen, sulfur, fluorine, chlorine, bromine, and hydrogen were removed $-2 \leq$ LogP $\leq$ 6 200 Da $\leq$ MW $\leq$ 600 Da 3D structures were generated by the LigPrep module in Schrödinger	ZINC Clean Lead

BRICS, breaking of retrosynthetically interesting chemical substructures; LogP, lipophilicity; MSTG, molecular substructure tree generative model.

F. The role of foundation models

Recent advances in AI have introduced powerful tools for scientific research, including specialized language models and multimodal systems. Among these, Galactica stands out as a specialized large language model developed to process and interpret complex scientific literature.[138] By training on an unprecedented corpus of 48 million research papers and technical documents, it achieves superior performance in tasks requiring deep domain knowledge compared with general-purpose models. Researchers can leverage Galactica to accelerate literature reviews, generate hypotheses, and even assist in technical writing, significantly reducing the time traditionally spent on manual information synthesis.

In molecular sciences, MoleculeSTM represents a significant leap forward by bridging the gap between chemical structures and their textual descriptions.[139] This multimodal approach enables researchers to query molecular databases using natural language while simultaneously analyzing structural data. Such capability proves invaluable in drug discovery, where understanding structure-property relationships is crucial. For instance, a researcher could ask, "Which molecules share similar solubility properties with aspirin?" and receive precise answers combining both chemical and textual data.

Protein engineering has similarly benefited from AI advancements through tools like ProtGPT2.[140] Unlike conventional methods limited to known protein structures, this model generates novel, biologically viable sequences by learning from the entire protein space. This opens new possibilities in designing enzymes for industrial applications or developing therapeutic proteins that might not exist in nature. The model's ability to propose functional sequences could dramatically accelerate projects in synthetic biology and biomedical research.

Molecular docking, a critical step in drug development, has been transformed by DiffDock's innovative approach. Traditional docking methods often struggle with computational intensity and accuracy, particularly for flexible molecules. DiffDock addresses these challenges through a diffusion-based method that predicts binding poses with remarkable speed and precision.[88] Although it does not calculate binding affinities—a task requiring complementary methods—its pose predictions provide researchers with reliable starting points for further analysis. This capability makes it particularly useful in virtual screening campaigns where thousands of compounds need evaluation.

Despite these advancements, conventional AI models often fail to fully capture the complexity of molecular data. GIT-Mol addresses this limitation by integrating multiple data types—molecular graphs, images, and text—into a unified framework.[141] This multimodal architecture allows for more comprehensive analysis, such as

Table 7
The list of studies that utilize the ChEMBL dataset

Method	Number	Preprocessing	Version
VGAE-MCTS	1,352,672	Graph structure representation of a molecule converted into a vector	ChEMBL[145]
Abbasi et al[52]	1,178,946	Tokenized the SMILES containing regular tokens and combined tokens	
DeepTarget	551,223 drug-protein pairs selected from 1,417,957 pairs	Binarization Removing negative samples Removing invalid molecules Removing samples that are unknown, uncommon, or contain isomer information	
CRTmaccs	1,710,676 of 2,136,186	Remove SMILES strings containing disconnected ions or fragments Remove salts, isotopes and charge neutralized Remove molecule with QED or QEPPI < 0.3	ChEMBL30
Gircha et al[163]	192,000	Max SMILES length of 200 Contain the atoms from the organic subset only (B, C, N, O, P, S, F, Cl, Br, I) Remove salt residuals	ChEMBL26
Zhang et al[156]	1.4 million were used in the pretrained model. Evaluated by 2.1 million molecules of ChEMBL30	-	ChEMBL22 used in a pretrained model ChEMBL30 was used for evaluation
Dong et al[159]	Over 1.5 million molecules	-	ChEMBL[160]
DeepLigBuilder	1 million	QED > 0.5 Conformation generated with RDKit	-

predicting reaction outcomes by considering both structural drawings and written procedures. Such integration could prove transformative in fields like organic chemistry, where traditional machine learning approaches struggle with the nuanced relationships between different data representations.

The combined application of these tools offers even greater potential. For example, ProtGPT2 could generate novel protein sequences that are then evaluated using DiffDock for binding potential, whereas Galactica provides insights from existing literature about similar proteins. MoleculeSTM and GIT-Mol could further enhance this pipeline by enabling cross-modal analysis of the resulting molecules. Together, these AI systems not only accelerate individual research tasks but also create new possibilities for interdisciplinary discovery, potentially leading to breakthroughs that would be difficult or impossible through traditional methods alone. Their continued development and integration promise to further blur the boundaries between computational prediction and experimental validation in scientific research.

IX. Datasets

This section examines the diverse datasets employed in molecule-generating studies. Table 5[44,49,52,60,61,67–69,75–77,80,81,84–86,89,96,142–178] catalogs various approaches alongside their corresponding datasets and evaluation metrics, highlighting commonly used resources across different methods. The following subsections introduce and analyze these datasets in detail.

A. ZINC

ZINC represents a comprehensive, publicly accessible database aggregating commercially available and annotated compounds.[142,147,157,179] This resource provides downloadable 2D and 3D molecular versions and features a web interface enabling rapid molecule lookup and analog search. The database has exhibited remarkable growth, expanding from fewer than 1 million compounds in 2005 to approximately 2 billion in 2020.[157]

Despite this extensive collection, most research applications utilize only a limited subset of molecules, typically selected through random sampling or specific property filtering. ZINC includes specialized subsets such as biogenic-like compounds and ZINC Clean Lead—a collection of synthetic compounds exhibiting Tanimoto similarity of 80% or greater to biogenic libraries. Table 6[52,156,159,161,174,179–182] summarizes the de novo drug design studies utilizing the ZINC dataset.

B. ChEMBL

ChEMBL functions as an extensive, open-access drug discovery database designed to capture medicinal chemistry data across the pharmaceutical research and development spectrum.[145,149,160,183] The current database contains comprehensive information about 2.4 million molecules, 1.6 million assays, 15,000 drugs, 48,000 indications, 15,000 targets, 6900 mechanisms, 2000 cells, 782 tissues, and their interconnections, including drug-protein relations. Table 7[52,145,156,159,160,163] documents the reviewed studies employing ChEMBL for molecule generation.

C. CASF

CASF provides high-quality protein-ligand interaction scoring data.[143] The DEVELOP framework constructed test sets comprising 188 and 143 examples for linker design and scaffold elaboration, respectively, from CASF-2016. Similarly, DiffLinker utilized CASF-2016, obtaining 309 examples after preprocessing, comparable with the approach by Imrie et al[110] Generating test sets from CASF offers advantages over ZINC-derived test splits, as CASF contains experimentally verified 3D conformations, whereas ZINC molecules rely on RDKit for 3D conformer generation.

D. PDBbind

PDBbind stands as the first dataset systematically annotating protein-ligand complexes from the Protein Data Bank with experimental binding data.[144] The latest version (v. 2020) contains 19,443 complexes, with 5316 comprising the refined set. DEVELOP constructed training-analogous test sets of 321 and 288 examples for linker design and scaffold elaboration from the PDBbind Refined Set (v. 2019).[144] DiffHopp utilized 19,378 protein-ligand complexes from PDBbind, filtered for quantitative estimate of

Table 8
Summary of studies that utilize MOSES with their preprocessing steps

Method	Number	Preprocessing
CMolGPT	1,760,739	-
Haroon et al[169]	1.9 million	Tokenization with position value
GRAPHAF	1.9 million	-
Mol-Zero-GAN	-	-
GMTransformer	1,936,962	Tokenization

drug-likeness (QED) values ≥0.3, defining Murko-Bemis scaffolds[184] for each ligand. TagMol employed the PDBbind refined 2017 dataset containing 4506 protein-ligand pairs, with ligands exceeding 32 heavy atoms trimmed by removing atoms with minimal neighboring bonds, and protein files preprocessed to remove hydrogen atoms.

E. BindingDB

BindingDB serves as a repository of experimental protein-ligand affinity data, currently containing 2.9 million data points for 1.2 million compounds and 9300 targets. Among reviewed studies, DeepTarget[148] combined BindingDB with ChEMBL drug-protein pairs to create an expanded training dataset.

F. DrugBank

DrugBank provides a richly annotated resource containing detailed information on approved and investigational drugs with comprehensive target and action data. In experimental applications, CRTmaccs utilized 10,756 curated drug SMILES strings from DrugBank for comparative dataset analysis.

G. iPPI-DB

iPPI-DB represents a manually curated database of protein-protein interaction modulators. CRTmaccs employed 2361 curated SMILES strings from iPPI-DB for comparative analysis in experimental evaluations.

H. PubChem

PubChem functions as a public repository for chemical substance information and biological activity data. TransAntivirus[44] obtained 2D representations of molecules from PubChem, implementing a 2-stage filtering process. Initial filtering removed SMILES strings containing disconnected ions or fragments and standardized compounds using RDKit to remove salts and isotopes while neutralizing charges, yielding 106,459,817 retained molecules.

The second filtration stage applied property constraints:

- Molecular weight (MW) between 100 and 900.
- Lipophilicity between −5 and 8.
- Synthetic accessibility score below 10.
- Fewer than 5 hydrogen bond donors (HBDs).
- Fewer than 10 hydrogen bond acceptors (HBAs).
- Topological polar surface area below 150.
- QED score of 0.3 or higher.

This secondary filtering yielded 79,156,024 molecules, with 30 million ultimately selected for the training dataset.

Table 9
Summary of studies that utilize QM9 with their preprocessing steps

Method	Number	Preprocessing
CDGS	133,885	Kekulized Converted to graph
GeoLDM	134K	-
GRAPHAF	134K	-
MDM	134K	-
MG2N2	133,885	Converted to graph
QGAN-HG	133,885	-

QGAN-HG, quantum generative adversarial network with a hybrid generator.

Table 10
Overview of datasets used in this study, including names and source URLs

Dataset	Source URL
ZINC	https://zinc.docking.org https://zinc15.docking.org https://zinc12.docking.org http://zinc5.docking.org/
ChEMBL	https://www.ebi.ac.uk/chembl/
CASF	https://www.pdbbind-plus.org.cn/casf
PDBbind	https://www.pdbbind-plus.org.cn/download
BindingDB	https://www.bindingdb.org/rwd/bind/index.jsp
DrugBank	https://go.drugbank.com/
iPPI-DB	https://ippidb.pasteur.fr/
PubChem	https://pubchem.ncbi.nlm.nih.gov/
ChemBridge (commercial)	https://chembridge.com/
MOSES	https://github.com/molecularsets/moses
GuacaMol	https://github.com/BenevolentAI/guacamol
QM9	https://doi.org/10.6084/m9.figshare.978904
GEOM	https://github.com/learningmatter-mit/geom
Pockets dataset	https://github.com/igashov/DiffLinker
OE62	https://doi.org/10.6084/m9.figshare.11689347
PDB	https://www.rcsb.org/
ProfKin	http://www.lilab-ecust.cn/profkin/
CrossDocked	https://github.com/gnina/models
DUD-E	https://dude.docking.org/
ExCAPE-DB	https://solr.ideaconsult.net/search/excape/

I. Commercial ChemBridge database

ChemBridge provides high-quality screening compounds and libraries for hit identification, offering over 1.3 million readily available small molecule screening compounds. Zhang et al[156] utilized approximately 700,000 molecules from ChemBridge for initial docking calculations to train target-specific scoring models, employing common datasets (ZINC and ChEMBL) for evaluation.

J. MOSES

MOSES offers a benchmarking platform standardizing the training and comparison of molecular generative models, introducing datasets based on ZINC Clean Leads.[163] Studies employing this resource typically require minimal preprocessing, as the dataset was specifically designed for molecular generation tasks. Table 8[169] summarizes the various applications of MOSES across the reviewed studies.

K. GuacaMol

GuacaMol provides a framework for benchmarking de novo molecular design models, featuring a dataset derived from the ChEMBL 24 database. Tree-invent[165] was trained on the GuacaMol dataset comprising 1.2 million molecules. The researchers selected 242 ring system classes covering 97% of all ring systems in GuacaMol as predefined ring nodes for structure generation. Molecules exceeding 600 MW were removed, establishing maximum atom counts of 46 for molecules and 34 for ring systems, with maximum ring and tree nodes set to 8 and 42, respectively. All molecules underwent neutralization and Kekule form conversion, resulting in

a final reference dataset of 672,945 molecules. Haroon et al[169] also employed GuacaMol, which encompasses 1.6 million molecules.

L. QM9

QM9 contains molecular structures and properties derived from quantum chemistry calculations for the first 134,000 molecules from the chemical universe GDB-17 database.[166,185] Table 9 summarizes methods utilizing QM9 in their development pipelines. As this dataset was specifically created for benchmarking purposes, molecules typically undergo minimal preprocessing.

M. GEOM

GEOM generates 37 million molecular conformations using advanced sampling and semiempirical density functional theory for over 450,000 molecules. DiffLinker constructed a dataset based on GEOM[168] molecules decomposed into 3 or more fragments with 1 or 2 connecting linkers. Two fragmentation techniques achieved these splits: an matched molecular pair analysis-based algorithm[186] and breaking of retrosynthetically interesting chemical substructures,[181] yielding 41,907 molecules and 285,142 fragmentations. GeoLDM also employs GEOM, focusing on larger organic compounds and selecting the 30 lowest energy conformations of each molecule for training. MDM utilizes GEOM, containing 290,000 molecules, averaging 46 atoms each.

N. Pockets dataset

The protein-ligand dataset curated by Schneuing et al[84] assessed DiffLinker's ability to generate valid linkers with protein pocket information from Binding MOAD.[187] Pocket definitions included amino acids with at least 1 atom within 6Å of any ligand atom, with all atoms in these residues constituting the pocket. Molecule splitting using an matched molecular pair analysis-based algorithm[186] generated 186,734 molecular fragments.

O. OE62

Several studies employ more specialized datasets:

- **OE62:** Westermayr et al[172] utilized OE62, containing 62,000 functional organic molecules, to generate optimal molecules for organic electronics applications.
- **PDB:** The Protein Data Bank archive of biological macromolecular structural data was employed by Mol-Zero-GAN[175] to retrieve 3D protein structures.
- **ProfKin:** This manually curated kinase-ligand database was used by Mol-Zero-GAN[175] to obtain the 3D structure of the AKT1 target protein.
- **CrossDocked:** This structure-based machine learning dataset contains 22.5 million ligand poses docked into similar binding pockets across the Protein Data Bank. Luo et al[77] filtered the dataset to exclude entries with binding pose root mean squared deviation (RMSD) greater than 1Å, creating a refined subset of 184,057 data points, clustered at 30% sequence identity with 100,000 protein-ligand pairs selected for training and 100 proteins from remaining clusters for testing.

P. Task-based datasets

Many studies employ small, specialized datasets for training auxiliary modules or fine-tuning models for specific tasks such as property prediction or target-specific design:

- Abbasi et al[52] utilized 5 small datasets (ADORA2A, KOR, JAK2, USP7, and bbbp) for training predictive modules.
- DNMG selected subsets from DUD-E[188] to fine-tune models for specific target proteins.
- CRTmaccs collected small bioactivity datasets for MDM2-p53 protein-protein interactions from ChEMBL for activity prediction model training.
- TransAntivirus employed 2 literature-derived datasets for model fine-tuning and case studies in nucleoside and non-nucleoside analog design.
- Subramanian et al[161] used patent-mined datasets (OPD and TKI) for property-specific model optimization.
- cMolGPT[162] utilized 3 ExCAPE-DB datasets (EGFR, HTR1A, and S1PR1) for target-specific activity prediction.
- Tree-invent assessed utility across multiple tasks using target-specific datasets (DRD2, ADAM17, and HTR1A).
- Haroon et al[169] obtained datasets from ExCAPE-DB and compiled small collections from BindingDB, ChEMBL, and PDB for target-specific design against AKT1 and CDK2.

Q. Discussion

Most generative approaches employ relatively limited molecular subsets with drug-like properties during training to generate novel drug-like compounds at inference time. For instance, methods using the ZINC dataset typically work with several hundred thousand molecules, despite ZINC20 containing nearly 2 billion compounds. Although a primary objective of molecular generation methods is navigating vast chemical spaces,[81,151] current approaches fail to explore even existing chemical space due to restricted training datasets.

This limitation stems from exclusive training on drug-like molecules to generate similar compounds. A more effective strategy would involve training on more extensive and diverse datasets to develop broader input space understanding while employing specialized loss functions, architectural layers, or task-specific modules to constrain generation toward drug-like chemical spaces. Additionally, current task-specific methods utilize highly limited datasets, further restricting their generalization capabilities.

This study leverages multiple publicly available datasets to ensure robust benchmarking and analysis. A comprehensive list of these datasets, along with their respective sources and access links, is provided in Table 10 for reproducibility and transparency.

X. Metrics

This section examines the key metrics used to evaluate molecular generation methods. These metrics fall into 2 broad categories: numerical values used directly for comparative analysis (such as validity, novelty, uniqueness, recovery rate, and SC_{rdkit}) and property-based metrics that require statistical processing (means, medians, or distributions) for meaningful comparison between generated molecules, training datasets, or reference collections.

This review categorizes evaluation metrics into 3 fundamental groups:

1. **Correctness metrics**: These evaluate the overall validity of generated molecular sets without considering individual molecular properties.
2. **Structure analysis-based metrics**: These assess properties of individual molecules related to their potential as drugs by analyzing molecular structure, chemical composition, weight, and atomic spatial arrangement.
3. **Pharmacology-based metrics**: These measure how drugs function within biological systems, including absorption,

Table 11
Overview of 12 molecular modeling software packages, highlighting key features, primary use cases, and limitations
Tools are categorized by functionality: conformer generation, molecular docking, deep learning architectures, and graph-based frameworks.

Package	Website	Key Features	Primary Purpose	Limitations
RDKit	https://www.rdkit.org	- Open-source cheminformatics toolkit - Generates 3D conformers from SMILES/2D structures - Uses distance geometry + MMFF94/UFF force field optimization - Integrated with Python (easy ML pipeline integration)	Small molecule conformer generation, cheminformatics, ML featurization	- Not always biologically relevant conformations - Limited handling of macromolecules - Default settings may need tuning for specific cases
Open Babel	https://openbabel.org	- Supports 300+ chemical file formats - Uses genetic algorithm for conformer generation - Command-line interface with Python bindings	File format conversion and basic conformer generation	- Less accurate than specialized tools - Limited conformer optimization options - Older codebase with less active development
AutoDock Vina	https://vina.scripps.edu	- Fast, open-source docking - Uses scoring function based on empirical data - Supports GPU acceleration (Vina-GPU fork)	Protein-ligand docking for virtual screening	- Rigid protein residues - Limited solvent effects modeling - Scoring function can be inaccurate for some targets
Glide (Schrödinger)	https://www.schrodinger.com/products/glide	Hierarchical docking with precision levels - Advanced scoring functions (GlideScore, XP) - Handles protein flexibility better than Vina	High-accuracy commercial docking	- Proprietary (expensive license) - Computationally intensive - Steep learning curve
GOLD	https://www.ccdc.cam.ac.uk/solutions/csd-discovery/Components/Gold	- Genetic algorithm-based docking - Excellent handling of metal centers - Customizable scoring functions	Flexible ligand docking with metalloproteins	- Commercial software - Slow for large virtual screens - Requires careful parameter tuning
SchNet	https://github.com/atomistic-machinelearning/SchNet	- Continuous-filter convolutional layers - Learns atomistic potential energies - Built on PyTorch	Molecular property prediction from 3D structures	- Limited to single molecule properties - Not designed for generation - Requires 3D input data
DimeNet++	https://github.com/klicperajo/dimenet	- Directional message-passing - Incorporates bond angles - State-of-the-art on QM9 dataset	Accurate quantum property prediction	- Computationally expensive - Large memory requirements - Complex architecture
SE(3)-Transformers	https://github.com/FabianFuchsML/se3-transformer-public	- SE(3)-equivariant attention layers - Handles 3D point clouds - Built with TensorFlow	3D molecular generation and optimization	- Difficult to train - Limited pretrained models - High VRAM requirements
OpenChem	https://github.com/Mariewelt/OpenChem	- SMILES-based RNN/LSTM generators - Property prediction models - PyTorch-based	Simple molecular generation starter kit	- Limited active development - Basic architecture options - No graph/3D support
PyTorch Geometric	https://pytorch-geometric.readthedocs.io	- Optimized GNN operations - Prebuilt molecular graph layers - Scalable mini-batching	Graph-based molecular DL	- Steep learning curve - Requires PyTorch knowledge - Some operations not GPU-optimized
Deep Graph Library	https://www.dgl.ai	- Multi-backend support (PyTorch/TensorFlow) - Efficient sparse operations - Built-in molecular datasets	Scalable graph neural networks	- Less chemistry-specific features - Documentation can be technical

LSTM, long short-term memory.

distribution, metabolism, and excretion characteristics, often employing molecular docking to predict binding affinity and stability.

A. *Correctness-based metrics*

1. *Validity*

Most generative models assess molecular validity, though definitions vary based on data representation and task objectives:

- Imrie et al[76] defined validity as the percentage of generated molecules containing starting substructure(s) and parsable by RDKit.
- Levy and Rector-Brooks[174] considered molecules valid when they are fully connected and satisfy basic valency constraints.
- Li et al[67] defined validity in terms of chemical stability and high receptor binding site affinity.
- Some studies report "validity without check," representing the percentage of valid molecules without post hoc chemical valency correction.[69,171]

2. Uniqueness

Uniqueness measures the proportion of distinct molecules within a generated set. Implementation varies across studies:

- Imrie et al[76] assess uniqueness on a per-example basis.
- Other methods calculate uniqueness across fixed sample sizes: 1000, 10,000, or 100,000 generated molecules.[49,68,81,159]

The chosen sample size significantly impacts reported results.

3. Novelty

Novelty represents the proportion of generated molecules absent from the training dataset, measuring the model's capacity to explore chemical space beyond its training examples.

4. Recovery rate

For transformative methods that modify input molecules (such as molecular fragmentation in linker design), the recovery rate serves as a primary evaluation metric. This measures the proportion of molecules successfully recovered during generation, computed by comparing generated molecules with corresponding ground truth structures.[76,86]

5. Diversity

Diversity (or internal diversity) assesses chemical variation within generated molecular sets using the following formula[189]:

$$IntDiv_p = 1 - \sqrt[p]{\frac{1}{|G|^2}\sum_{m1,m2 \in G} T(m_1, m_2)^p} \quad (6)$$

where T is the Tanimoto similarity.

This metric effectively identifies mode collapse—a common failure where models produce limited sample variety by ignoring portions of chemical space. Higher values indicate greater diversity.[163]

Although this approach is widely implemented,[44,49,52,67,77,80,81,85,175] alternatives include the following:

- Maximum mean discrepancy calculated using Morgan fingerprints and USRCAT[190] for 2D and 3D chemical space.[75]
- Tanimoto coefficient similarity distribution analysis.[156]

External diversity, measuring similarity between generated and original molecules, additionally indicates model generalization capacity.[52]

Other correctness metrics include the following:

- **Reconstruction**: The GRAPHAF method defines reconstruction as the percentage of molecules reproducible from latent vectors.[171]
- **Element distribution**: Compares atomic frequency distributions between generated and training molecules.[172]
- **Learning analysis**: TagMol[167] evaluates loss function, predicted binding energy, and FD score during training.
- **Precision and recall**: Nonparametric manifold representations of real and generated data estimate distribution coverage and sample quality.[75]

B. Structure analysis based on metrics

1. Molecular filters

Several methods[49,76,80,86] employ filtering approaches as evaluation metrics:

- Lipinski's "Rule of Five."
- Exclusion of pan assay interference compounds substructures.[182]
- Synthetic accessibility score thresholds.
- "Rule of Three" criteria.[191]

Models often apply these same filters during training data preparation, potentially biasing generated molecules toward passing these criteria.

2. Shape and color similarity score SC_{rdkit}

This metric employs 2 RDKit functions based on methods by Putta et al[192] and Landrum et al[193]:

- Color similarity scores 3D conformer overlap based on pharmacophoric features.
- Shape similarity performs volumetric comparison between conformers.

Each measure produces scores between 0 (no match) and 1 (perfect match), which are averaged. Scores above 0.7 indicate good matches, whereas scores exceeding 0.9 suggest near-perfect matches.[110] This metric is particularly valuable when similarity between original and generated molecules is relevant for evaluation.[76,86]

3. Root mean squared deviation

RMSD measures similarity between original and generated molecules.[76,86] Li et al[75] employed RMSD for conformational quality assessment, measuring heavy atom positions before and after optimization with MMFF94s.[194]

4. Synthetic accessibility

SAscore[195] evaluates drug-like molecules for virtual screening, calculated as the sum of fragment scores and complexity penalties. The fragment score reflects the frequency of ECFP4 fragments[196] in the PubChem database, whereas the complexity penalty considers structural elements like aromatic rings, stereocenters, macrocycles, and molecular size.[197]

5. Molecular weight

More than 10 works reviewed in this study use a molecule's weight as a molecular property.

6. Number of atoms

This metric computes the number of atoms in the molecule—another simple metric used by Wang et al.[61]

7. Similarity to the nearest neighbor

Similarity to the nearest neighbor calculates the average Tanimoto similarity between molecules in the generated set and their nearest neighbors in the reference dataset:

$$SNN(G,R) = \frac{1}{|G|}\sum_{m_G \in G} \max_{m_R \in R} T(m_G, m_R) \quad (7)$$

This metric functions as a precision measure—if generated molecules diverge significantly from the reference set manifold, nearest neighbor similarity will be low.[163]

8. Fragment similarity

Frag computes distance between generated and reference molecular sets[181] using cosine similarity: as the number of times a substructure f appears in molecules from set A, and F denotes a set of fragments that appear in either G or R, then, the metric is defined as a cosine similarity:

$$Frag(G,R)=\frac{\sum_{f\in F}\left[c_f(G).c_f(R)\right]}{\sqrt{\sum_{f\in F}c_f^2(G)}\sqrt{\sum_{f\in F}c_f^2(R)}} \quad (8)$$

where c_f (A) represents the frequency of substructure f in set A, and F denotes the set of fragments appearing in either G or R.

High values indicate similar fragment distributions between sets, whereas low values suggest over- or under-representation of certain fragments in the generated set.[163]

9. Scaffold similarity

Similar to fragment similarity, but comparing Bemis-Murcko scaffold frequencies[184] instead of fragments:

$$Scaff(G,R)=\frac{\sum_{s\in S}[c_s(G).c_s(R)]}{\sqrt{\sum_{s\in S}c_s^2(G)}\sqrt{\sum_{s\in S}c_s^2(R)}} \quad (9)$$

This metric reveals scaffold distribution similarities between generated and reference datasets, with low values indicating underrepresentation of certain chemotypes.[163]

Additional structure-based metrics:

- **Topological polar surface area**: summed contributions of polar atoms and attached hydrogens, indicating membrane permeability potential.[169]
- **HBDs/HBAs**: distribution analysis[75] or range assessment.[162]
- **Neighborhood subgraph pairwise distance kernel**: mean maximum discrepancy between generated and test molecules, considering node and edge features.[198]
- **Connectivity**: assessment of molecular graph connectedness.[85]
- **Number of rings**: critical property for drug design applications.[86]
- **Stability**: proportion of atoms with correct valency (atom stability) or molecules with all stable atoms (molecule stability).[89,170]
- **Rotatable bonds**: count of rotatable bonds.[75]
- **Normalized principal moment of inertia ratios (NPR1 and NPR2)**: visualization of molecular shape diversity.[199]
- **Bond geometry**: distribution analysis of bond lengths, angles, and torsions for conformer quality assessment.[75]
- **Advanced structure check**: software-based structural error detection.[159,200]

C. Pharmacology-based metrics

1. KL divergence

GuacaMol[96] implements Kullback-Leibler divergence[201] to measure how well-generated molecular distributions approximate training set distributions across physicochemical descriptors. The approach captures both distribution fidelity and diversity by requiring generated molecules to match training set property distributions.[52,60,161,165]

The metric aggregates KL divergence across multiple physicochemical descriptors[96]:

$$S=\frac{1}{k}\sum_{i}^{k}\exp\left(-D_{KL,i}\right) \quad (10)$$

where k represents the number of descriptors (typically 9).

2. Fréchet ChemNet distance

Fréchet ChemNet distance[202] measures similarity between generated and training molecule distributions using ChemNet, a neural network trained for biological activity prediction. The metric calculates the Fréchet distance between activation means and covariances from ChemNet's penultimate layer for both reference and generated sets. Lower values indicate greater distribution similarity.[96]

This metric is widely implemented in recent studies.[44,60,69,80,81,151,161,165]

3. Quantitative estimate of drug likeness

QED,[203] used in over 20 reviewed studies, fits desirability functions to property distributions across 771 marketed oral drugs for 8 key properties:

- MW
- Lipophilicity
- HBDs.
- HBAs.
- Polar surface area.
- Rotatable bonds.
- Aromatic rings.
- Undesirable substructure alerts.

The metric assigns higher scores when compound properties match common patterns in marketed drugs, combining individual property desirabilities into a single geometric mean score.[204]

4. Affinity or docking score

Binding affinity and docking scores measure protein binding capability, obtained from databases or computational tools like DeepPurpose,[205] Glide,[206] Vina,[207–209] or Smina.[210]

Evaluation approaches include[77,81,85,148,156,158]:

- Distribution, mean, or median analysis.
- High-affinity percentage assessment.
- Comparison of model-generated and tool-calculated docking scores.

Zhang et al[156] introduced an enrichment factor comparing molecular generation to conventional docking:

$$EF_x=\frac{N_g}{N_e} \quad (11)$$

$N_e$where N_g represents generated molecules with Vina scores below threshold x, and N_e represents compounds with docking scores below x in an enumerated library.

Additional pharmacology-based metrics:

- **QEPPI**: Quantitative estimate of protein-protein interaction targeting drug-likeness.[211]
- **Conditional generation properties**: Mean absolute error for molecular properties in conditional generation.[89,170]
- **Electronic properties**: application-specific properties for specialized molecules.[172]
- **Solvent accessible surface areas**: measurement of molecule-solvent contact area.[212]
- **pIC50**: potency quantification for specific biological target inhibition.
- **Docking pose generation**: comparison between known active compounds and similar generated molecules.[156]
- **Natural product-likeness score**: Bayesian measure of similarity to natural product structural space.[213]

D. Molecular modeling software

To empower readers in selecting appropriate tools for molecular modeling tasks, we systematically evaluated 12 key software packages covering conformer generation (RDKit, Open Babel), molecular docking (AutoDock Vina, Glide, GOLD), deep learning architectures (SchNet, DimeNet++, SE(3)-Transformers), and graph-based frameworks (PyTorch Geometric, Deep Graph Library). Our comparison reveals distinct specialization areas: RDKit excels in cheminformatics pipelines due to its Python integration and MMFF94 force field optimization, whereas SE(3)-Transformers lead in 3D-aware generation through equivariant neural networks, albeit with substantial GPU memory requirements. For virtual screening, AutoDock Vina offers the best speed-accuracy tradeoff among open-source options, although commercial tools like Glide provide superior flexible-receptor handling. Critical limitations were identified across tools, including RDKit's occasional biologically irrelevant conformers and SE(3)-Transformers' training complexity. This curated comparison (Table 11) serves as a practical guide for method selection based on project requirements, computational resources, and desired accuracy levels.

E. Discussion

A significant challenge in evaluating generative models stems from the absence of ground truth information for novel molecules. Consequently, evaluation relies on either simple metrics or general statistical measures like averages and distributions across generated sets. To address this limitation, researchers could develop methods with high generation controllability, enabling the intentional exclusion of known molecules from training sets for subsequent generation during evaluation. This approach would provide ground truth information for a comprehensive assessment. Task redefinition and integrated datasets incorporating diverse data types could further enhance controllability. Alternatively, when generating held-out molecules proves exceptionally challenging, developing metrics that quantify similarity between generated molecules and held-out references could provide valuable evaluation alternatives.

As mentioned, a key challenge in evaluating novel AI-generated molecules lies in the absence of ground truth data for validation. As future work, learnable evaluation metrics could be developed based on property predictors, trained not only on existing datasets but also iteratively updated with new data from AI-generated candidates. These predictors would estimate critical properties (eg, binding affinity, toxicity, or synthesizability) and function as dynamic, adaptive proxies for ground truth. For practical adoption, the predictors must prioritize explainability: interpretable models (eg, linear models with feature importance, graph-based explainers for GNNs) or post hoc explanation tools (eg, Shapley Additive Explanations values) could reveal the structural features driving predictions (eg, "this molecule's low score stems from its reactive functional group"). Over time, as additional AI-generated molecules are validated (either computationally or experimentally), the predictors would improve, establishing a feedback loop that refines both the generative model and its evaluation framework. This approach would transition evaluation from static benchmarks to a self-improving system, where metrics evolve alongside the generative pipeline while preserving transparency in decision-making.

XI. Summary and outlook

This review has categorized drug design methods according to molecular representation types (1D, 2D, and 3D) while examining relevant datasets and evaluation metrics. This discussion addresses 3 key questions that emerge from our comprehensive review.

A. Comparing input and output representations

The choice of molecular representation significantly impacts generative model performance. When evaluating representation suitability, validity preservation emerges as a critical factor, ensuring generated structures correspond to valid molecules according to chemical rules and constraints.

Linear representations like SMILES, although widely used, impose significant constraints on generative approaches due to their strict syntactic rules. Similarly, 3D representations face validity challenges as the generated atomic coordinates must satisfy multiple spatial and chemical constraints. In contrast, 2D graph representations offer better suitability for generative output, balancing expressiveness with constraint management, although they necessarily sacrifice some molecular information.

For input representation, the ability to extract and encode comprehensive structural information becomes paramount. 3D representations provide the most informative approach, capturing spatial relationships crucial for understanding molecular function. Recognizing the limited 3D information availability in real-world scenarios, Liu et al[73] developed GraphMVP, which enhances 2D representations through semisupervised learning between 3D and 2D views, demonstrating superior molecular property encoding compared with alternative approaches.

B. Comparing generative approaches

The comparative analysis of generative approaches encompasses several critical dimensions:

C. Model assumptions

Different generative frameworks operate under distinct assumptions:

- VAE-based models assume latent variables following prior distributions (typically Gaussian), which can be limiting.
- Denoising diffusion models assume Gaussian transition distributions in forward diffusion processes, enabling efficient sampling and reparameterization but potentially constraining model flexibility.
- All model types assume sufficient network capacity to accurately model data distributions—insufficient capacity limits the ability to capture distribution complexity.

Mode collapse presents a significant challenge, particularly prevalent in GAN-based models, limiting molecular novelty and diversity. Analysis of implementation frequencies (Figs. 3 and 4) reveals a stronger preference for VAE and diffusion-based models over GAN approaches.

D. Model reliability

Each generative framework demonstrates distinct reliability characteristics:

- VAEs excel at learning smooth, continuous latent representations, generating diverse samples, and handling incomplete data, but may struggle with fine details and complex structural features.

- GANs capture intricate details and structural knowledge effectively but suffer from training instability, mode collapse, and evaluation challenges.
- Denoising diffusion models generate high-quality, diverse samples with stable training dynamics and excellent detail preservation, although at higher computational cost and slower sampling speeds compared with alternatives.

E. Auxiliary knowledge integration

Incorporating auxiliary knowledge enhances model performance across 2 primary categories:

- Structural knowledge integration.
- Goal-directed optimization.

These knowledge types can be incorporated either explicitly (direct integration into model architecture) or implicitly (commonly through RL techniques that provide directional signals for generating molecules with desired properties).

F. Comparing datasets and evaluation measures

Generative models require extensive, diverse datasets to develop strong generalization capabilities. Although global datasets provide valuable pretraining foundations, specialized datasets become essential for targeted applications such as designing drugs for specific protein targets or cell lines.

The evaluation landscape spans correctness metrics, structure-based analyses, and pharmacological assessments, each providing different perspectives on model performance. Robust evaluation requires multifaceted assessment across these dimensions, with metrics carefully selected based on intended application domains.

The challenge of ground truth absence for novel molecules remains significant, with potential solutions including controlled generation of known molecules intentionally excluded from training or development of sophisticated similarity metrics for comparing generated molecules with reference structures.

AI methods provide transformative advantages over traditional bioinformatics approaches. For generative chemistry, deep learning models can explore vast, uncharted chemical space and propose novel scaffolds beyond human intuition or rule-based systems while simultaneously optimizing multiple molecular properties—a task that would require sequential optimization using conventional methods. In structure-based design, AI overcomes limitations of classical docking by predicting protein flexibility and allosteric pockets through hybrid deep learning-molecular dynamics approaches and enables polypharmacology design through learned target embeddings. For property prediction, AI excels at extrapolating to novel chemotypes and modeling complex endpoints by integrating heterogeneous data types while providing crucial uncertainty estimates that traditional QSAR lacks. Also, traditional bioinformatics methods may still be preferable due to their interpretability or lower computational costs.

The integration of generative AI models into traditional medicinal chemistry workflows represents a paradigm shift in drug discovery, combining computational exploration with empirical medicinal chemistry expertise. Modern template-based approaches like SCENT[214] demonstrate how reaction-aware generation can directly output synthesizable candidates that align with medicinal chemists' design cycles while incorporating real-world constraints such as synthetic route feasibility and building block availability. These systems increasingly incorporate retrosynthesis planning tools[97] and on-the-fly absorption, distribution, metabolism, excretion, and toxicity prediction to filter proposals against established medicinal chemistry rules, creating a more seamless transition from in silico design to experimental validation.

The most effective implementations employ human-AI collaborative frameworks where medicinal chemists maintain strategic oversight while leveraging AI's exploration capabilities. Interactive platforms now allow chemists to steer generation through structural constraints and MPO,[215] with explainable reaction trajectories providing chemically interpretable rationales for AI proposals. This addresses a key adoption barrier—the traditional distrust of black-box systems in medicinal chemistry teams. Validation cycles have similarly evolved, with automated synthesis feasibility scoring combining computational retrosynthesis tools with empirical reaction success data,[216] and active learning systems incorporating experimental results to refine subsequent generations.

However, significant integration challenges remain. Current template systems still cover less than 50% of reactions commonly used in medicinal chemistry,[92] whereas reliable in silico patentability assessment continues to elude the field. The encoding of implicit medicinal chemistry knowledge, such as stability rules for specific functional groups, into generative systems remains incomplete. Emerging best practices suggest hybrid systems that combine generative AI with rule-based filters and QSAR models,[96] standardized workflows connecting to electronic lab notebooks, and continuous learning from newly synthesized compounds show particular promise.

The field appears to be moving toward closed-loop drug design ecosystems where generative models propose candidates, experimental teams validate them, and results feed back to improve subsequent generations. This integration is particularly valuable in lead optimization phases where synthetic feasibility becomes paramount. Future developments will need to focus on expanding reaction coverage, improving knowledge capture from medicinal chemistry teams, and enhancing interoperability with pharmaceutical R&D infrastructure to fully realize the potential of this AI-medchem integration.

G. Practical applications

Generative molecular design powered by AI is revolutionizing biotechnology and pharmaceutical development, with groundbreaking real-world applications already demonstrating its transformative potential. One of the most significant breakthroughs comes from Insilico Medicine, which developed Rentosertib, the first fully AI-designed drug for idiopathic pulmonary fibrosis. This achievement showcases AI's ability to not only identify novel biological targets but also design effective therapeutic compounds, compressing the traditional drug discovery timeline from years to just 18 months. The drug has already shown promising results in phase 2 trials, improving lung function in patients where current treatments only slow disease progression.

Beyond small-molecule drugs, companies like 310.ai are pushing boundaries in protein engineering through generative AI models. Their MP4 system can create entirely novel protein sequences from simple text prompts, bypassing the need for natural templates. In lab tests, 84% of AI-designed proteins were successfully expressed (far exceeding expert predictions of 20%–30%). Some even showed high thermostability, a key trait for drug development. This accelerates the creation of proteins for therapeutics, enzymes, and biomaterials without relying on existing natural sequences. Many disease-related proteins are considered "undruggable" because they lack binding sites for traditional drugs. AI can design entirely new protein-based drugs that interact with these targets, opening doors for previously untreatable

diseases. 310.ai designed a functional protein responsive to ATP, with potential applications in heart disease and stroke treatment. Medicine's AI-generated TNIK inhibitor (INS018_055) for idiopathic pulmonary fibrosis is in phase 2 trials, showcasing a 30-month discovery timeline and utilizing PandaOmics and Chemistry42.

In early 2020, UK-based Exscientia, in collaboration with Japan's Sumitomo Dainippon Pharma, revealed that its AI-designed drug candidate, DSP-1181, had advanced to phase 1 clinical trials for obsessive-compulsive disorder. It was developed with the aid of AI, reportedly reaching phase 1 trials 5 times faster than traditional methods. This rapid advancement highlights the potential of AI in drug discovery. Although the provided search results mention the drug and its target, they do not detail the reasons for the phase 1 discontinuation or the lessons learned. Further investigation is needed to understand the specific challenges encountered and how these insights have informed Exscientia's subsequent research efforts.

H. Empirical verification

The validation of computationally generated molecules involves a multistep approach combining bioinformatic methods and experimental techniques. The process begins with comprehensive computational validation, where consensus scoring strategies combine molecular docking (eg, AutoDock Vina and Glide) with all-atom molecular dynamics simulations (eg, AMBER andGROMACS) to assess binding pose stability (via RMSD/root-mean-square-fluctuation analysis), thermodynamic properties (molecular mechanics/Poisson-Boltzmann surface area/Generalized Born surface area), and conformational dynamics (principal component analysis). These predictions are further reinforced through multimodal verification, where generative model outputs are cross-validated using QSAR predictions, physics-based free-energy calculations (free energy perturbation and thermodynamic integration), and machine learning models trained on orthogonal datasets (eg, ChEMBL vs PubChem) to minimize bias. To close the design-validate loop, experimental-informatics feedback mechanisms are implemented—early-stage assay data (eg, binding affinity measurements) are iteratively incorporated via active learning or Bayesian optimization to refine subsequent computational cycles.

Following in silico prioritization, experimental validation proceeds through 3 critical tiers: (1) structural characterization via high-throughput screening, nuclear magnetic resonance spectroscopy, and high-resolution mass spectrometry to confirm molecular identity and purity; (2) functional assessment through in vitro binding assays (SPR, ELISA) and cell-based activity screens, followed by in vivo pharmacokinetic/pharmacodynamic studies; and (3) target engagement verification using X-ray crystallography or cryo-electron microscopy for binding pose resolution, supplemented by mutagenesis studies to probe critical interactions. Finally, computational property predictions (eg, solubility, absorption, distribution, metabolism, excretion, and toxicity) are empirically confirmed through standardized wet-lab assays—equilibrium solubility measurements, hepatocyte stability tests, and in vivo bioavailability studies—to bridge the gap between prediction and physiological relevance. This integrated framework ensures both the reliability of computational designs and their translational potential.

I. Hallucinations and inaccuracies in computational drug design

The concern regarding AI-generated inaccuracies and hallucinations in computational drug design is highly valid and warrants careful consideration. AI models, particularly those used for de novo molecule generation, can indeed produce chemically invalid structures or biologically implausible candidates due to limitations in training data, architectural biases, or the inherent stochasticity of generative processes. For instance, a model might propose a molecule with unrealistic bond configurations or unstable ring systems that violate fundamental chemical principles. Even more critically, some AI-generated molecules may demonstrate promising in silico binding scores but ultimately fail during experimental validation due to poor solubility, toxicity, or synthetic infeasibility. This discrepancy between computational predictions and real-world performance underscores the risks of relying solely on AI for drug candidate design.

To mitigate these risks, the field has adopted multitiered evaluation strategies. Initial filters often include rule-based checks for chemical validity, such as ensuring proper valency, ring stability, and adherence to synthetic accessibility metrics like SAscore or Lipinski's Rule of Five. Tools like RDKit are routinely used to flag and discard invalid structures early in the pipeline. Beyond these basic checks, in silico validation methods such as molecular docking and dynamics simulations play a pivotal role in assessing the functional viability of AI-generated molecules. For example, a candidate predicted to inhibit a target protein might undergo docking studies with AutoDock or Schrödinger to evaluate binding affinity, followed by molecular dynamics simulations in GROMACS to probe thermodynamic stability over time. These steps help identify compounds that, although chemically valid, may lack the necessary interactions or stability to function as viable drugs.

Looking ahead, emerging techniques aim to further reduce reliance on post hoc corrections by embedding safeguards directly into the generative process. Hybrid approaches that combine AI with physics-based methods, such as force field optimization or fragment-based growing, can steer molecule generation toward more realistic chemical space. Uncertainty quantification methods, including Bayesian Neural Networks, provide confidence estimates for model outputs, allowing researchers to prioritize high-probability candidates for further study. Explainable AI tools like Shapley Additive Explanations analysis or attention mapping also offer insights into model decision-making, helping chemists understand why a particular molecule was proposed. Finally, iterative human-in-the-loop frameworks, where AI-generated candidates are refined through feedback from medicinal chemists, ensure that domain expertise remains central to the design process.

In summary, although AI's propensity for hallucinations and inaccuracies poses real challenges in drug design, the integration of rigorous validation pipelines, hybrid modeling approaches, and human oversight has significantly mitigated these risks. The goal is not to replace traditional methods but to augment them with AI's ability to explore vast chemical spaces efficiently. By maintaining a critical balance between innovation and validation, the field can harness AI's potential while safeguarding against its pitfalls. I would be curious to hear your perspective on whether these evolving solutions adequately address the concerns you raised.

Abbreviations

1D, one-dimensional; 2D, 2-dimensional; 3D, 3-dimensional; AI, artificial intelligence; BPE, byte pair encoding; DFPM, denoising diffusion probabilistic model; GAN, generative adversarial network; GNN, graph neural network; GPS, graph property synthesis; GPT, generative pretraining transformer; GPU, graphics processing unit; GraphMVP, graph multiview pretraining; HBA, hydrogen bond acceptor; HBD, hydrogen bond donor; L-Net, Ligand Neural Network; LogP, Lipophilicity; MCTS, Monte Carlo tree search; MDM, molecular diffusion model; MOSES, molecular

sets; MPNN, message passing neural network; MPO, multiparameter optimization; MW, molecular weight; NLP, natural language processing; NPR, normalized principal ratio; PDB, Protein Data Bank; QED, quantitative estimate of drug-likeness; QSAR, quantitative structure-activity relationship; RL, reinforcement learning; RMSD, root mean squared deviation; RNN, recurrent neural network; SD-VAE, syntax-directed variational autoencoder; SELFIES, Self-Referencing Embedded Strings; SMILES, Simplified Molecular Input Line Entry System; VAE, variational autoencoder; VGAE, variational graph autoencoder.

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